

```
In [2]: #Advanced EDA for Healthcare Domain
```

```
In [3]: import pandas as pd
```

```
In [4]: heart = pd.read_csv(r'C:\Users\SID Balurkar\Desktop\Full Stack Data Science\Classwork\Nov 2025\26th, 27th- Advanced EDA\heart.csv')
heart
```

```
Out[4]:
```

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
0	63	1	3	145	233	1	0	150	0	2.3	0	0	1	1
1	37	1	2	130	250	0	1	187	0	3.5	0	0	2	1
2	41	0	1	130	204	0	0	172	0	1.4	2	0	2	1
3	56	1	1	120	236	0	1	178	0	0.8	2	0	2	1
4	57	0	0	120	354	0	1	163	1	0.6	2	0	2	1
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
298	57	0	0	140	241	0	1	123	1	0.2	1	0	3	0
299	45	1	3	110	264	0	1	132	0	1.2	1	0	3	0
300	68	1	0	144	193	1	1	141	0	3.4	1	2	3	0
301	57	1	0	130	131	0	1	115	1	1.2	1	1	3	0
302	57	0	1	130	236	0	0	174	0	0.0	1	1	2	0

303 rows × 14 columns

```
In [5]: import numpy as np
```

```
In [6]: type(heart)
```

```
Out[6]: pandas.core.frame.DataFrame
```

```
In [7]: len(heart)
```

Out[7]: 303

```
In [8]: heart.columns
```

```
Out[8]: Index(['age', 'sex', 'cp', 'trestbps', 'chol', 'fbs', 'restecg', 'thalach',  
              'exang', 'oldpeak', 'slope', 'ca', 'thal', 'target'],  
              dtype='object')
```

```
In [10]: heart.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 303 entries, 0 to 302  
Data columns (total 14 columns):  
#   Column      Non-Null Count  Dtype  
---  ---  
0   age         303 non-null    int64  
1   sex         303 non-null    int64  
2   cp          303 non-null    int64  
3   trestbps    303 non-null    int64  
4   chol        303 non-null    int64  
5   fbs         303 non-null    int64  
6   restecg     303 non-null    int64  
7   thalach     303 non-null    int64  
8   exang       303 non-null    int64  
9   oldpeak     303 non-null    float64  
10  slope       303 non-null    int64  
11  ca          303 non-null    int64  
12  thal        303 non-null    int64  
13  target      303 non-null    int64  
dtypes: float64(1), int64(13)  
memory usage: 33.3 KB
```

```
In [11]: heart.head()
```

```
Out[11]:
```

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
0	63	1	3	145	233	1	0	150	0	2.3	0	0	1	1
1	37	1	2	130	250	0	1	187	0	3.5	0	0	2	1
2	41	0	1	130	204	0	0	172	0	1.4	2	0	2	1
3	56	1	1	120	236	0	1	178	0	0.8	2	0	2	1
4	57	0	0	120	354	0	1	163	1	0.6	2	0	2	1

```
In [12]: heart.tail()
```

```
Out[12]:
```

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
298	57	0	0	140	241	0	1	123	1	0.2	1	0	3	0
299	45	1	3	110	264	0	1	132	0	1.2	1	0	3	0
300	68	1	0	144	193	1	1	141	0	3.4	1	2	3	0
301	57	1	0	130	131	0	1	115	1	1.2	1	1	3	0
302	57	0	1	130	236	0	0	174	0	0.0	1	1	2	0

```
In [13]: heart.shape
```

```
Out[13]: (303, 14)
```

```
In [14]: heart.describe()
```

Out[14]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
count	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000
mean	54.366337	0.683168	0.966997	131.623762	246.264026	0.148515	0.528053	149.646865	0.326733	1.039604
std	9.082101	0.466011	1.032052	17.538143	51.830751	0.356198	0.525860	22.905161	0.469794	1.161075
min	29.000000	0.000000	0.000000	94.000000	126.000000	0.000000	0.000000	71.000000	0.000000	0.000000
25%	47.500000	0.000000	0.000000	120.000000	211.000000	0.000000	0.000000	133.500000	0.000000	0.000000
50%	55.000000	1.000000	1.000000	130.000000	240.000000	0.000000	1.000000	153.000000	0.000000	0.800000
75%	61.000000	1.000000	2.000000	140.000000	274.500000	0.000000	1.000000	166.000000	1.000000	1.600000
max	77.000000	1.000000	3.000000	200.000000	564.000000	1.000000	2.000000	202.000000	1.000000	6.200000



In [15]:

```
heart.describe().transpose()
```

Out[15]:

	count	mean	std	min	25%	50%	75%	max
age	303.0	54.366337	9.082101	29.0	47.5	55.0	61.0	77.0
sex	303.0	0.683168	0.466011	0.0	0.0	1.0	1.0	1.0
cp	303.0	0.966997	1.032052	0.0	0.0	1.0	2.0	3.0
trestbps	303.0	131.623762	17.538143	94.0	120.0	130.0	140.0	200.0
chol	303.0	246.264026	51.830751	126.0	211.0	240.0	274.5	564.0
fbs	303.0	0.148515	0.356198	0.0	0.0	0.0	0.0	1.0
restecg	303.0	0.528053	0.525860	0.0	0.0	1.0	1.0	2.0
thalach	303.0	149.646865	22.905161	71.0	133.5	153.0	166.0	202.0
exang	303.0	0.326733	0.469794	0.0	0.0	0.0	1.0	1.0
oldpeak	303.0	1.039604	1.161075	0.0	0.0	0.8	1.6	6.2
slope	303.0	1.399340	0.616226	0.0	1.0	1.0	2.0	2.0
ca	303.0	0.729373	1.022606	0.0	0.0	0.0	1.0	4.0
thal	303.0	2.313531	0.612277	0.0	2.0	2.0	3.0	3.0
target	303.0	0.544554	0.498835	0.0	0.0	1.0	1.0	1.0

In [16]: heart.info()

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 303 entries, 0 to 302
Data columns (total 14 columns):
 #   Column      Non-Null Count  Dtype
---  -
 0   age         303 non-null   int64
 1   sex         303 non-null   int64
 2   cp          303 non-null   int64
 3   trestbps    303 non-null   int64
 4   chol        303 non-null   int64
 5   fbs         303 non-null   int64
 6   restecg     303 non-null   int64
 7   thalach     303 non-null   int64
 8   exang       303 non-null   int64
 9   oldpeak     303 non-null   float64
10   slope       303 non-null   int64
11   ca          303 non-null   int64
12   thal        303 non-null   int64
13   target      303 non-null   int64
dtypes: float64(1), int64(13)
memory usage: 33.3 KB

```

```
In [18]: heart['target'].nunique()
```

```
Out[18]: 2
```

```
In [19]: heart['target'].unique()
```

```
Out[19]: array([1, 0])
```

```
In [22]: heart['target'].value_counts()
```

```

Out[22]: target
1      165
0      138
Name: count, dtype: int64

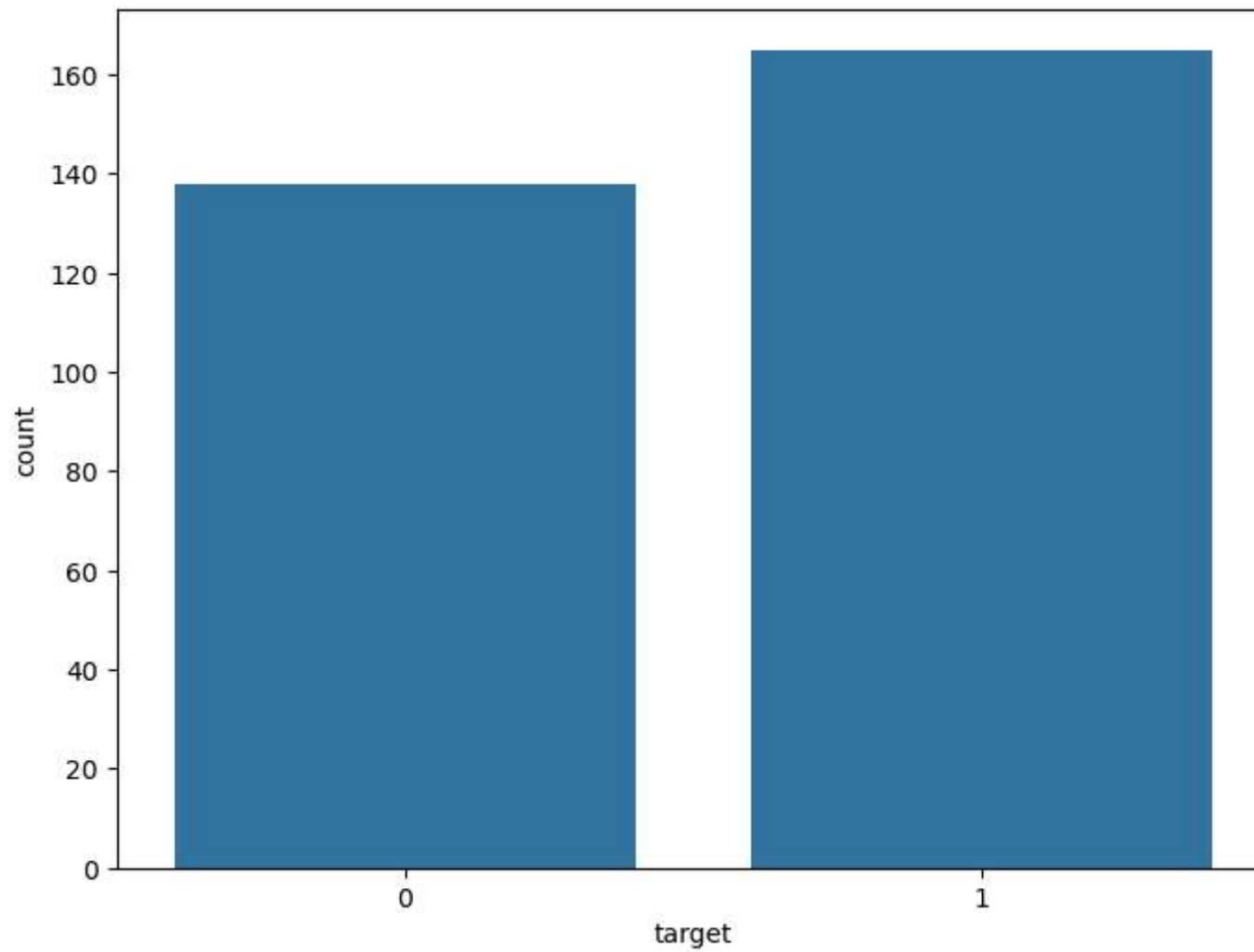
```

```

In [23]: import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')

```

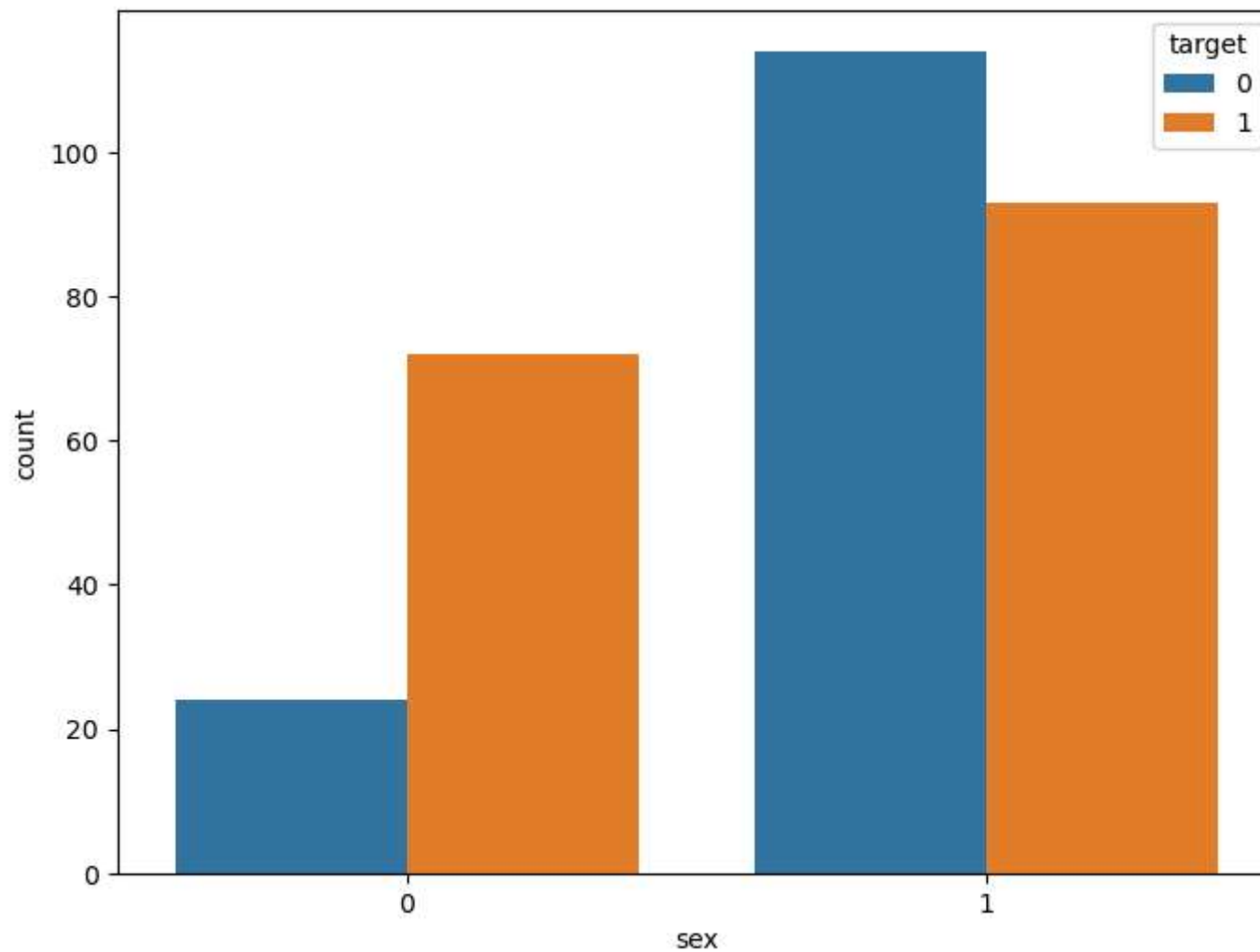
```
In [27]: f, ax = plt.subplots(figsize=(8,6))  
ax = sns.countplot(x="target", data = heart)  
plt.show()
```



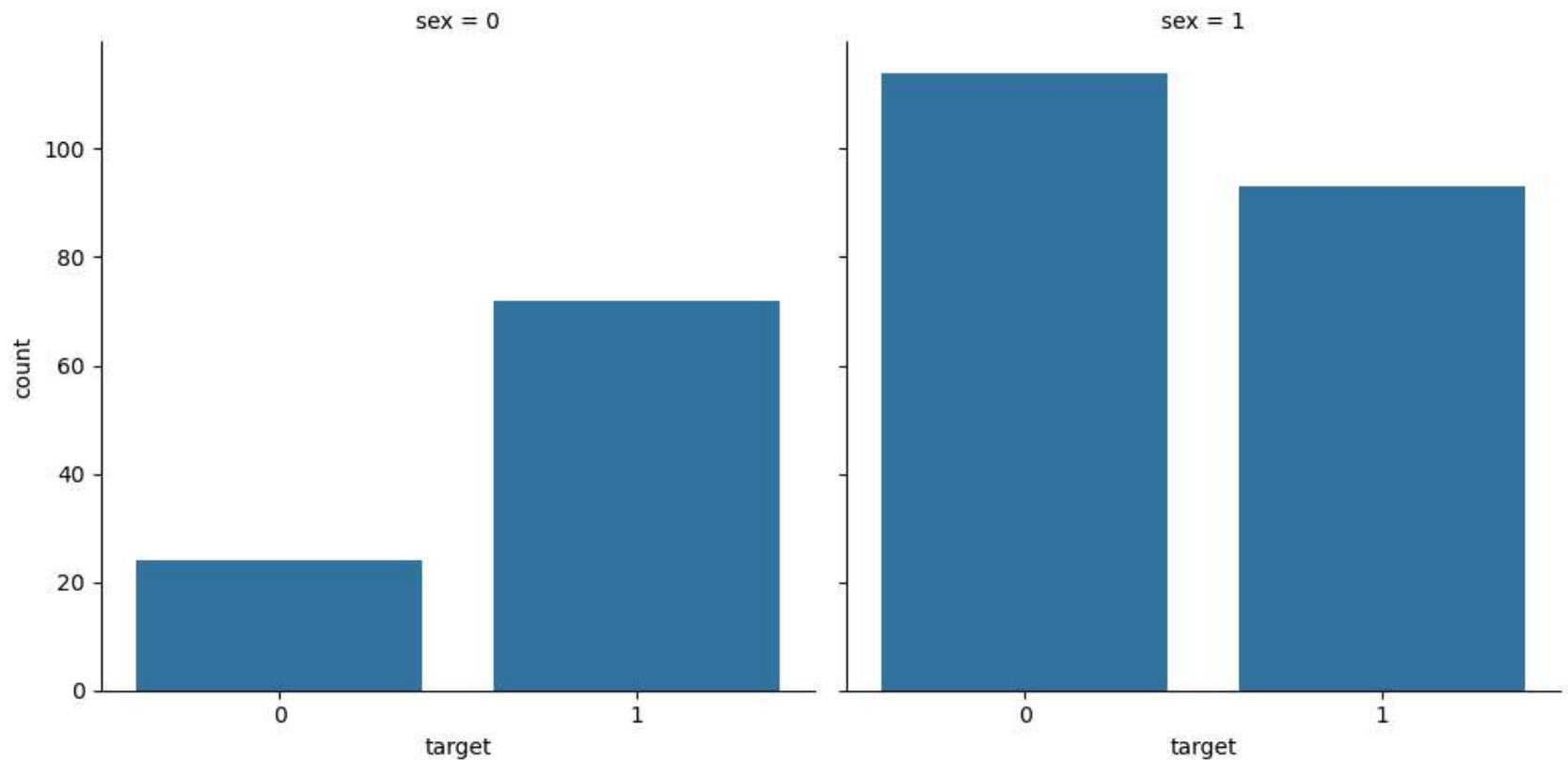
```
In [29]: heart.groupby('sex')['target'].value_counts()
```

```
Out[29]: sex target
         0    1      72
         0    0      24
         1    0     114
         1    1      93
        Name: count, dtype: int64
```

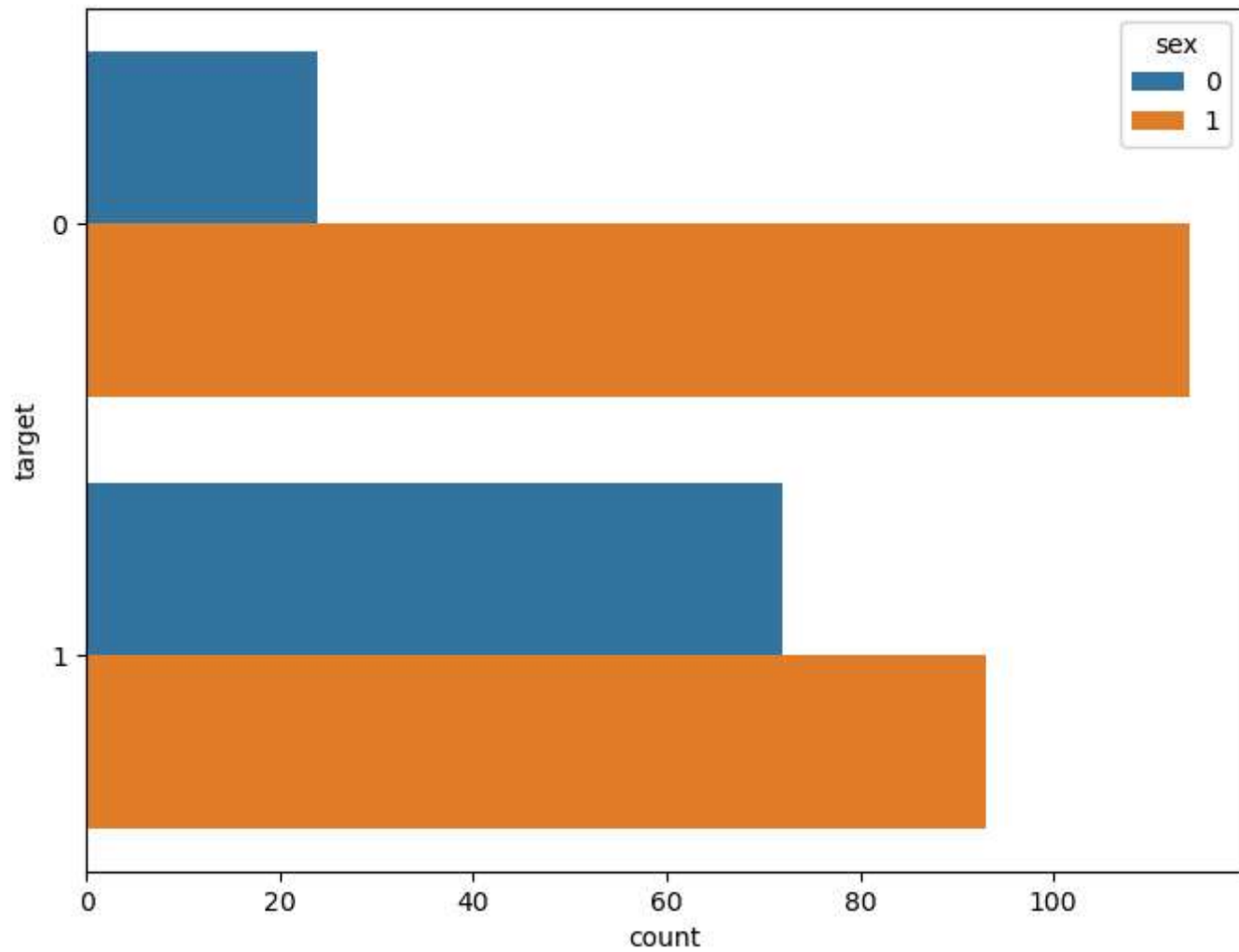
```
In [30]: f, ax = plt.subplots(figsize=(8,6))
         ax = sns.countplot(x="sex", hue="target", data = heart)
         plt.show()
```



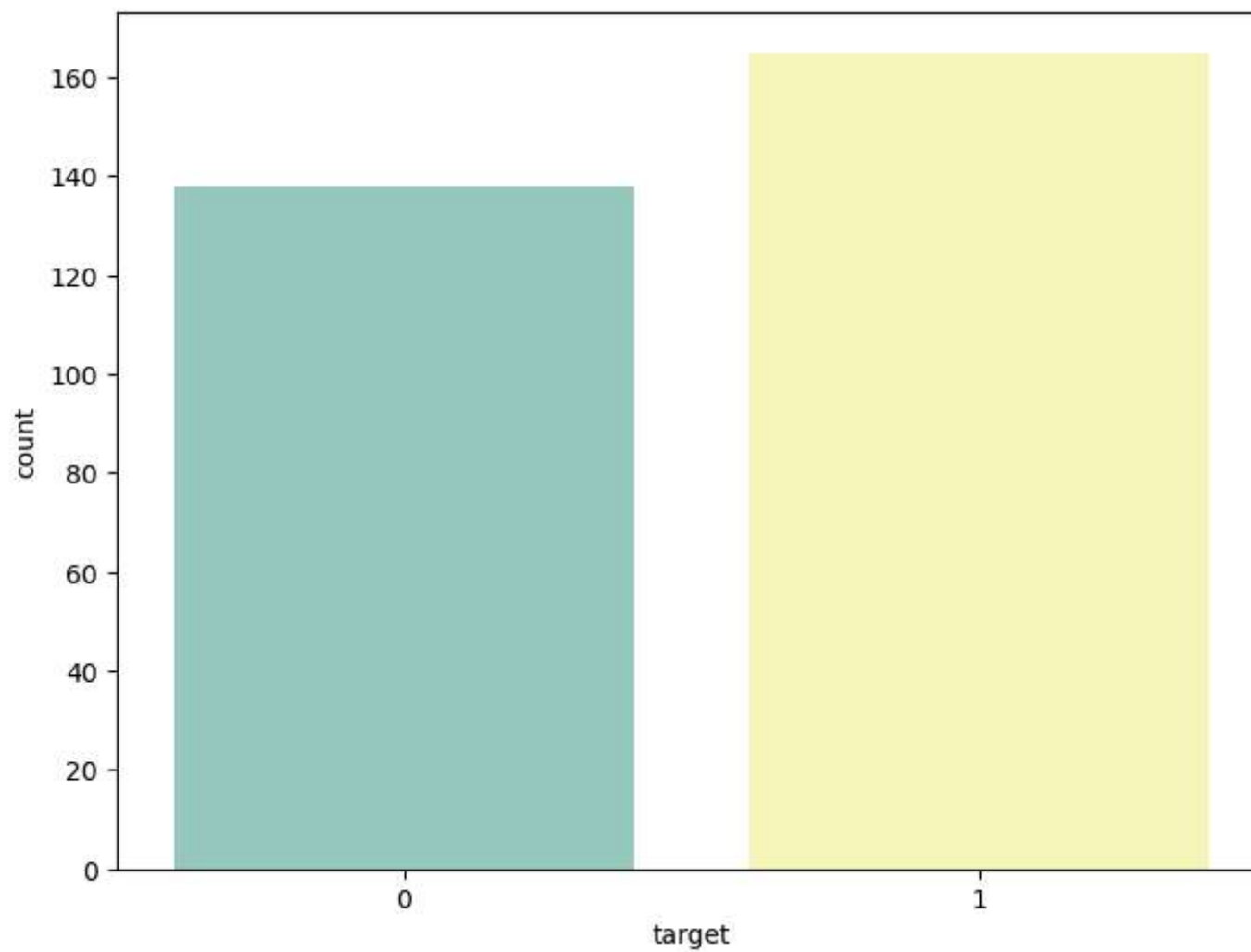

```
In [34]: ax = sns.catplot(x='target', col='sex', data = heart, kind = "count", height = 5, aspect = 1)
```



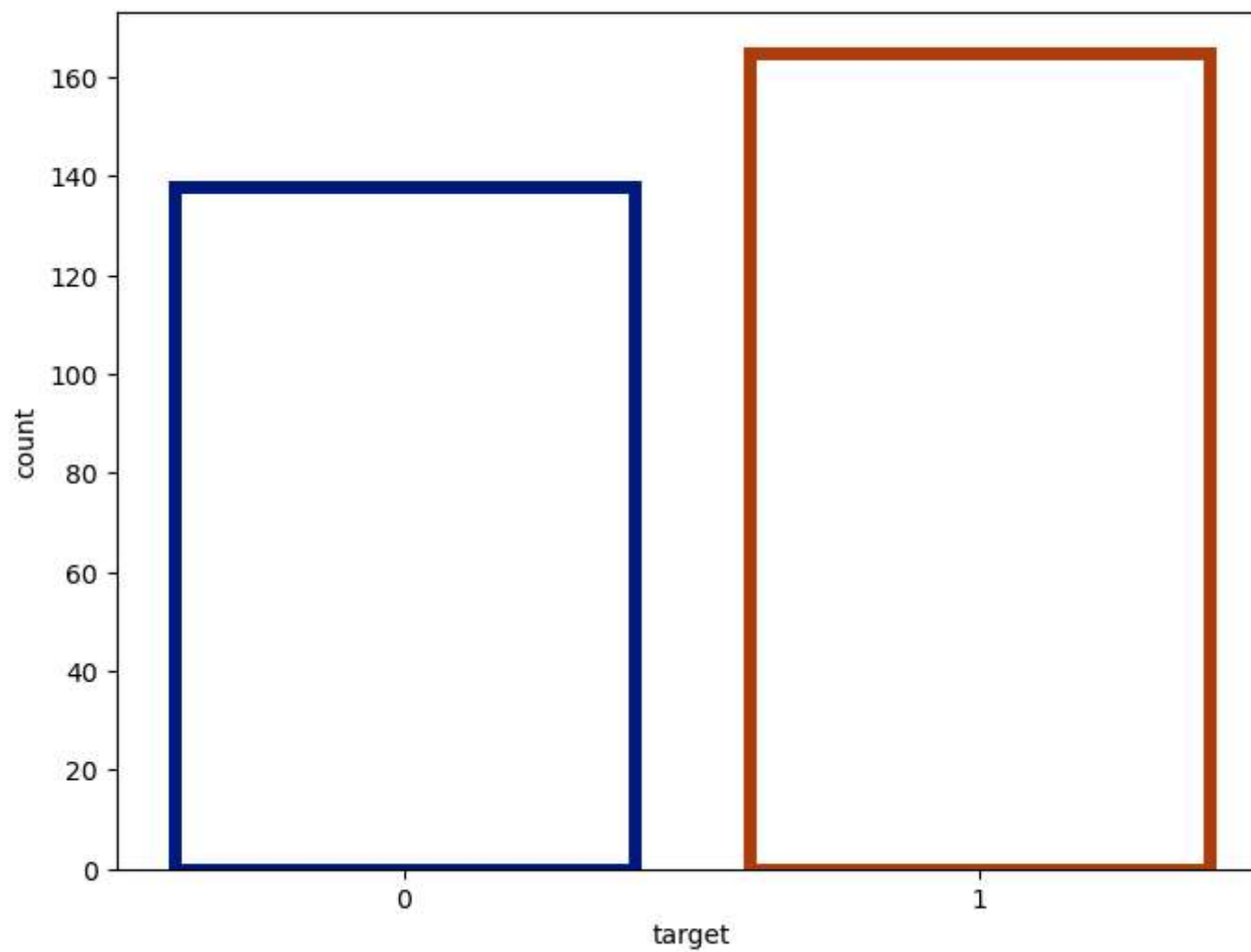
```
In [35]: f, ax = plt.subplots(figsize=(8,6))  
ax = sns.countplot(y="target", hue="sex", data = heart)  
plt.show()
```



```
In [39]: f, ax = plt.subplots(figsize=(8, 6))
ax = sns.countplot(x="target", data = heart, palette = "Set3")
plt.show()
```



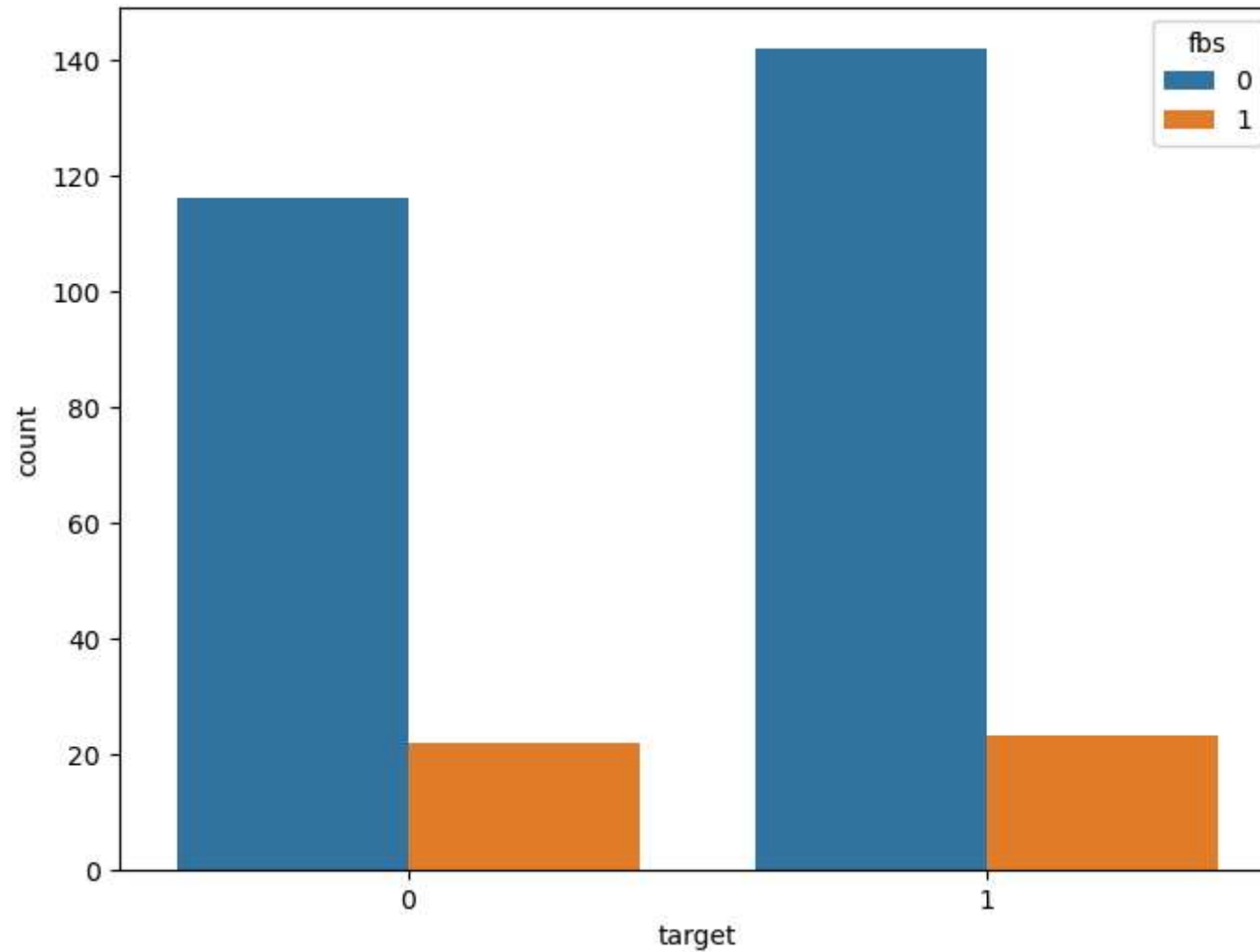
```
In [43]: f, ax = plt.subplots(figsize=(8, 6))
ax = sns.countplot(x="target", data = heart, facecolor = (0,0,0,0), linewidth = 5, edgecolor=sns.color_palette("dark", 2))
plt.show()
```



```
In [45]: heart.groupby('fbs')['target'].value_counts()
```

```
Out[45]: fbs  target
0      1      142
        0      116
1      1       23
        0       22
Name: count, dtype: int64
```

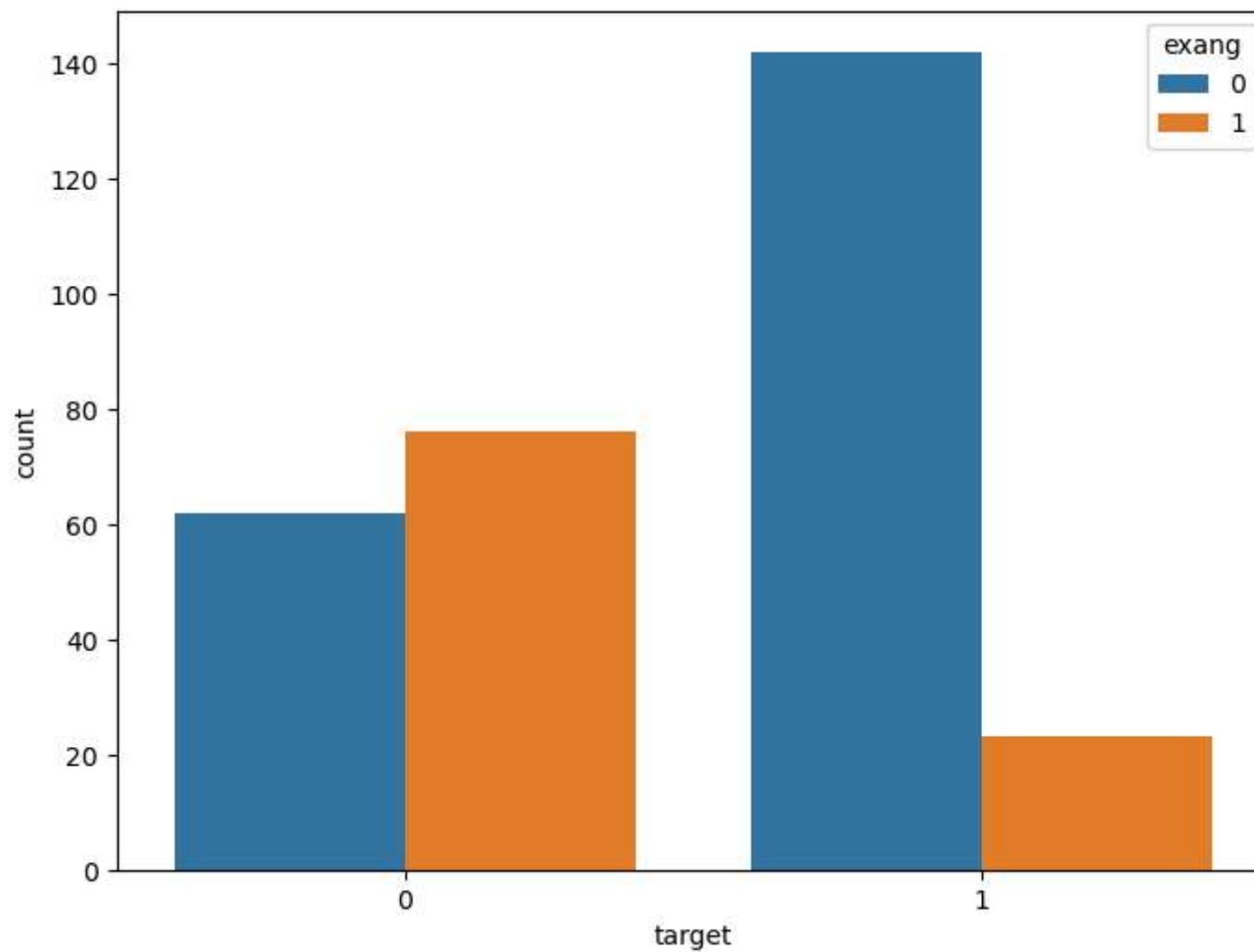
```
In [44]: f, ax = plt.subplots(figsize=(8, 6))  
ax = sns.countplot(x="target", hue = "fbs", data = heart)  
plt.show()
```



```
In [46]: heart.groupby('exang')['target'].value_counts()
```

```
Out[46]: exang  target
         0      1      142
         0      0       62
         1      0       76
         1      1       23
Name: count, dtype: int64
```

```
In [47]: f, ax = plt.subplots(figsize=(8, 6))
ax = sns.countplot(x="target", hue = "exang", data = heart)
plt.show()
```



```
In [48]: correlation = heart.corr()
```

```
In [49]: correlation['target'].sort_values(ascending=False)
```

```
Out[49]: target      1.000000  
cp          0.433798  
thalach     0.421741  
slope       0.345877  
restecg     0.137230  
fbs         -0.028046  
chol        -0.085239  
trestbps    -0.144931  
age         -0.225439  
sex         -0.280937  
thal        -0.344029  
ca          -0.391724  
oldpeak     -0.430696  
exang       -0.436757  
Name: target, dtype: float64
```

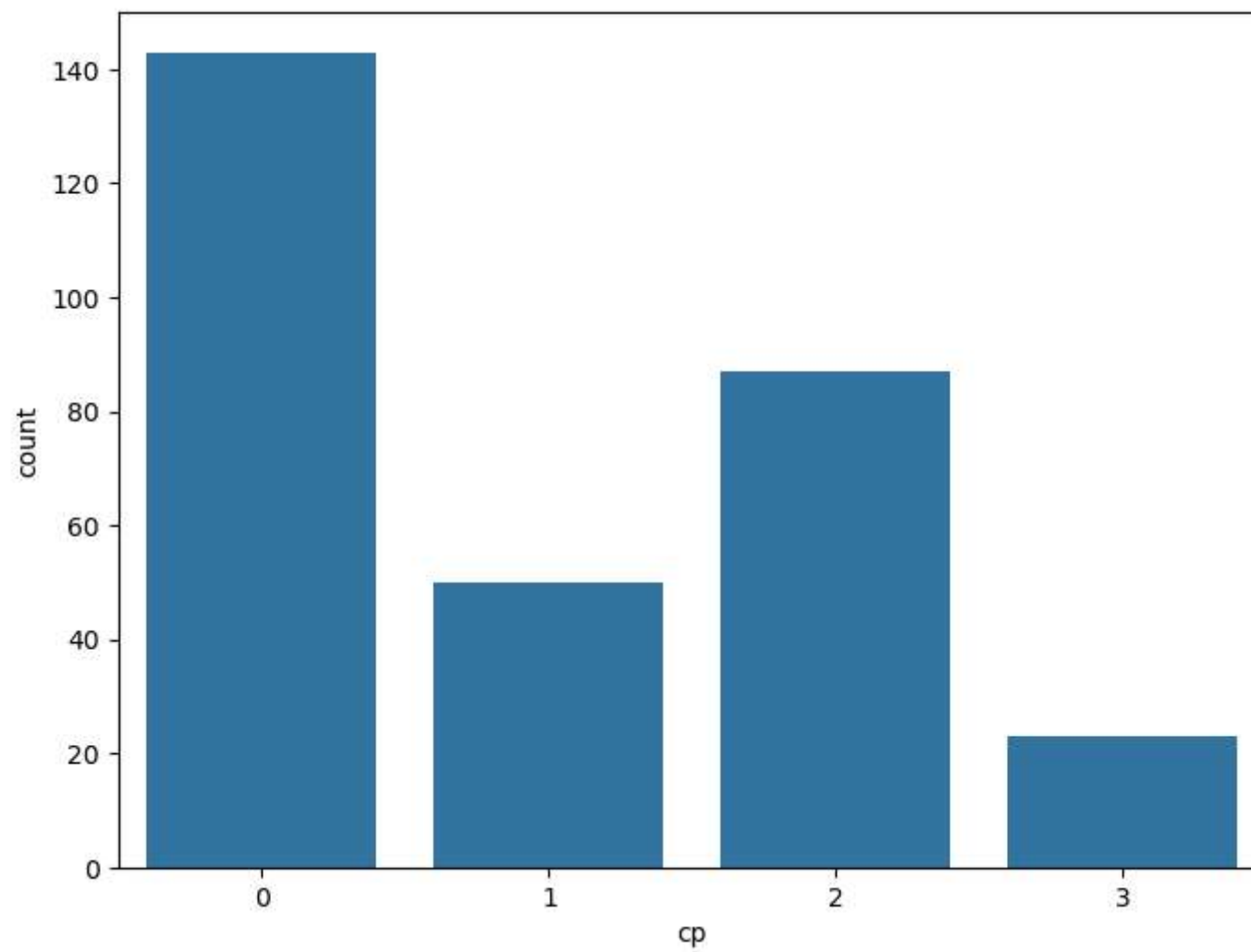
```
In [50]: heart['cp'].nunique()
```

```
Out[50]: 4
```

```
In [51]: heart['cp'].value_counts()
```

```
Out[51]: cp  
0      143  
2       87  
1       50  
3       23  
Name: count, dtype: int64
```

```
In [53]: f, ax = plt.subplots(figsize=(8, 6))  
ax = sns.countplot(x="cp", data = heart)  
plt.show()
```

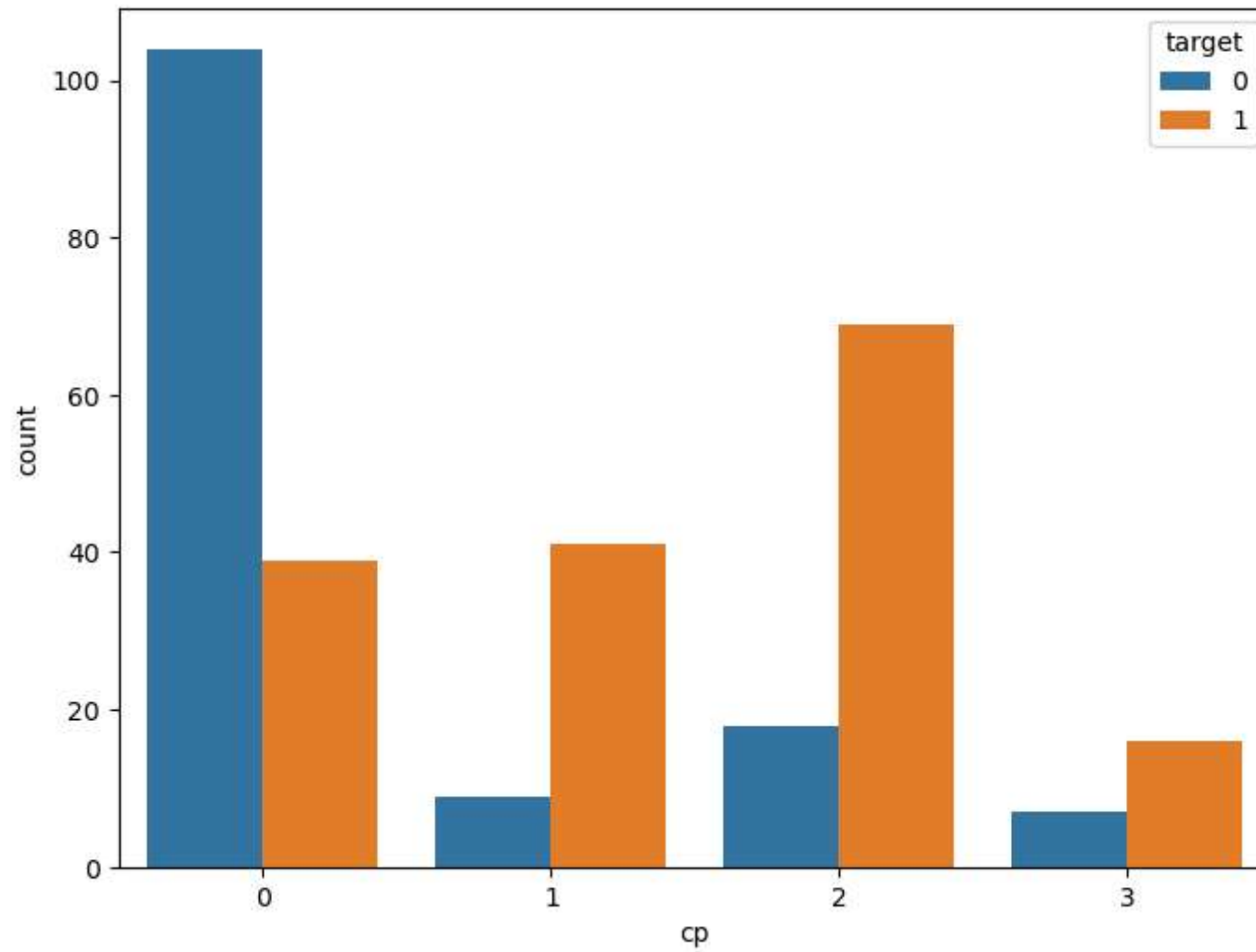


```
In [54]: heart.groupby('cp')['target'].value_counts()
```

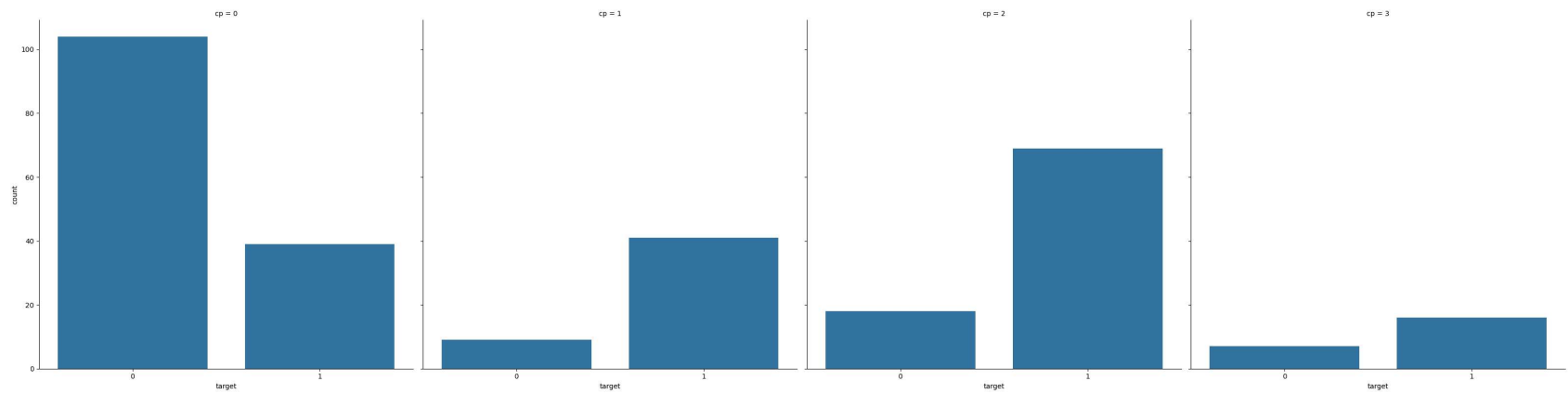


```
Out[54]: cp  target
          0    104
          1    39
          1    41
          0     9
          2    69
          0    18
          3    16
          0     7
          Name: count, dtype: int64
```

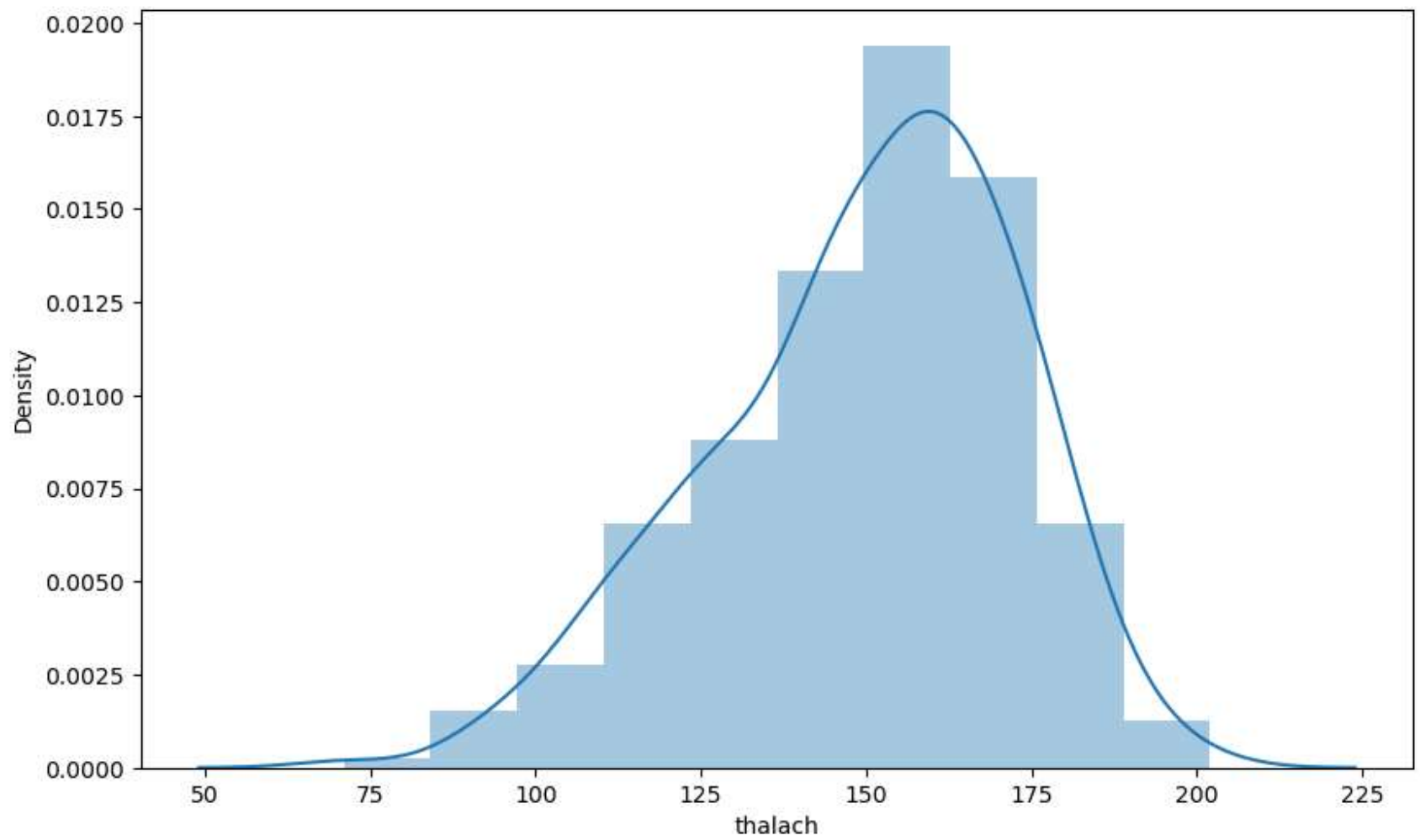
```
In [55]: f, ax = plt.subplots(figsize=(8, 6))
          ax = sns.countplot(x="cp", hue = "target", data = heart)
          plt.show()
```



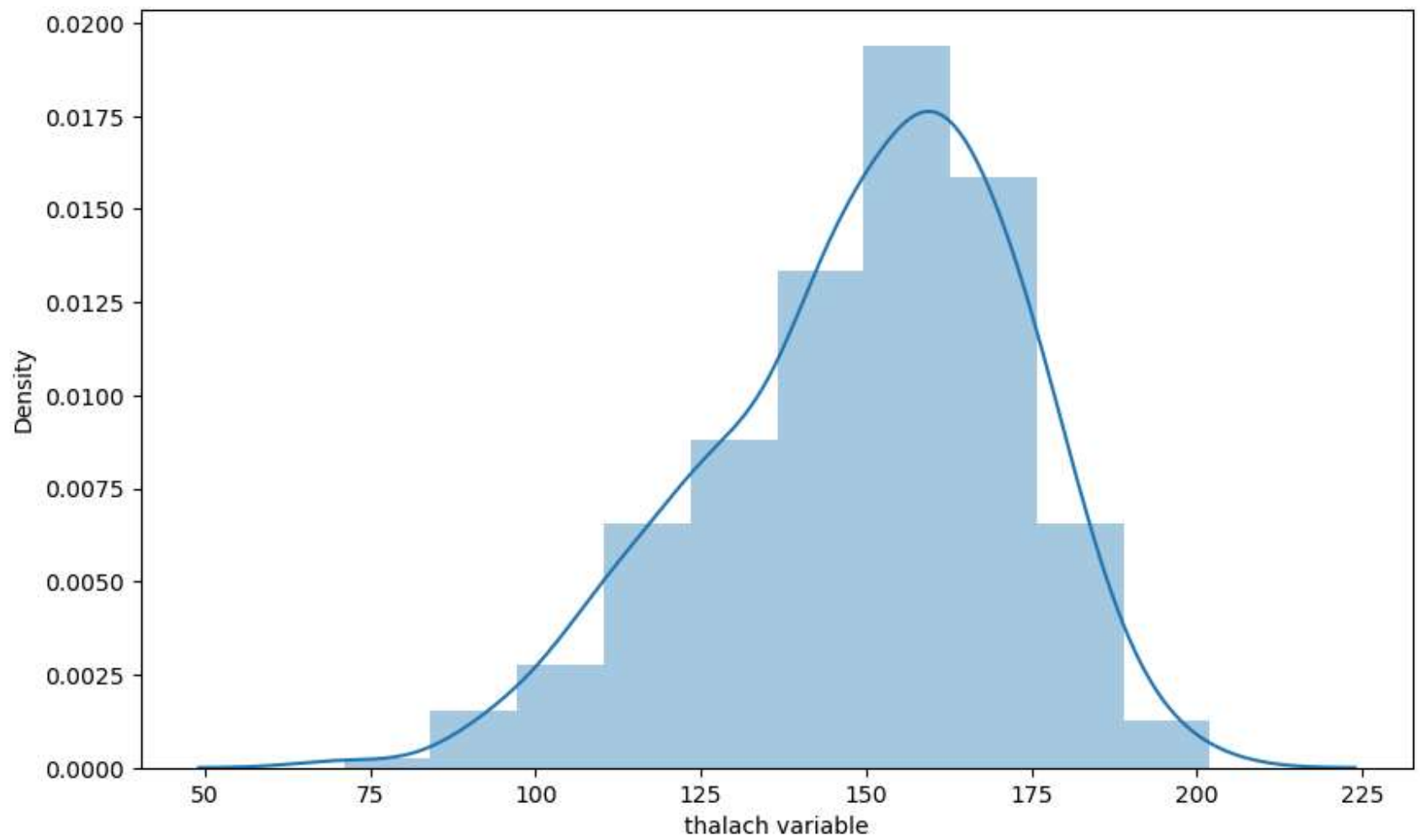
```
In [56]: ax = sns.catplot(x="target", col="cp", data = heart, kind = "count", height = 8, aspect = 1)
```



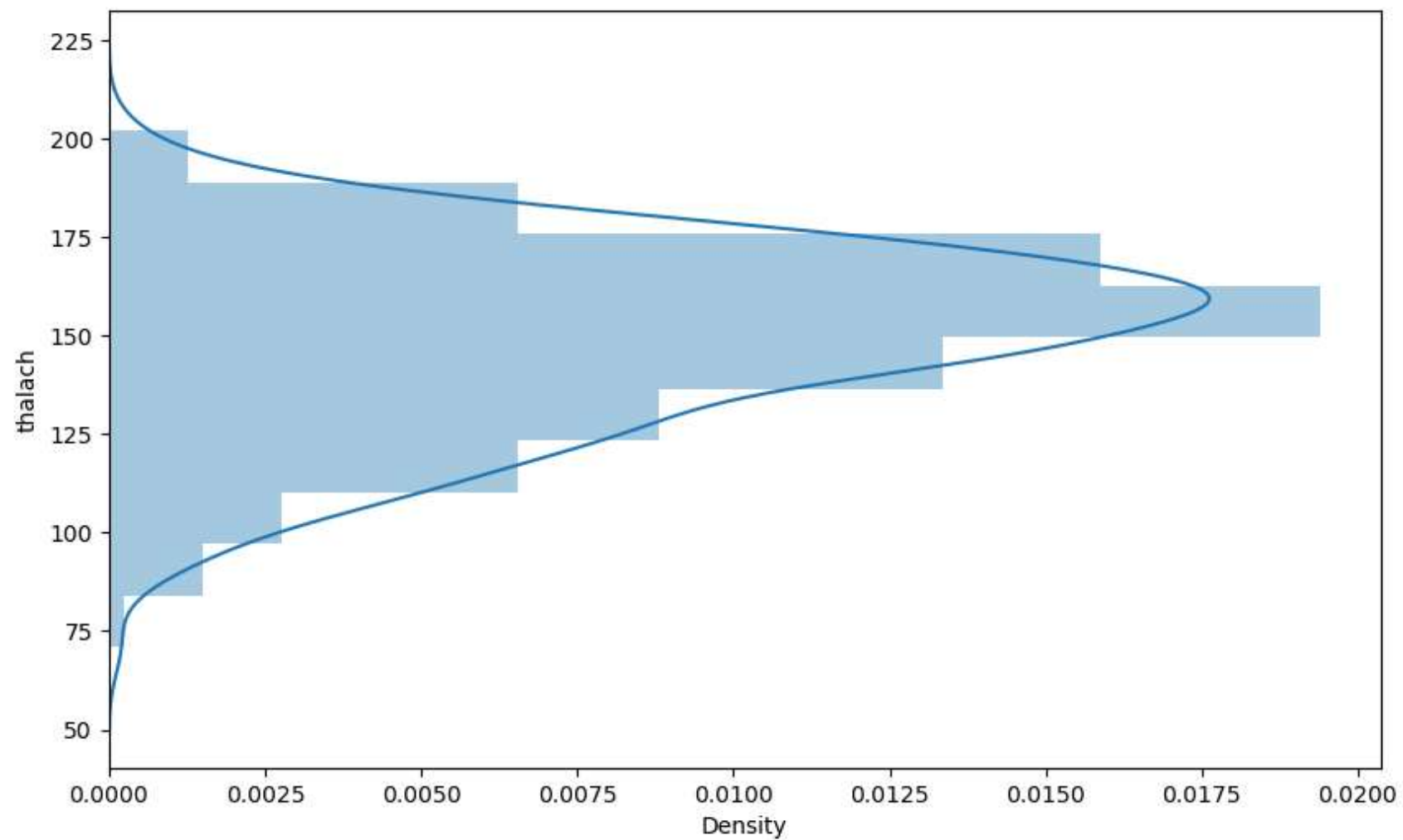
```
In [59]: f, ax = plt.subplots(figsize = (10,6))
x = heart['thalach']
ax = sns.distplot(x, bins = 10)
plt.show()
```



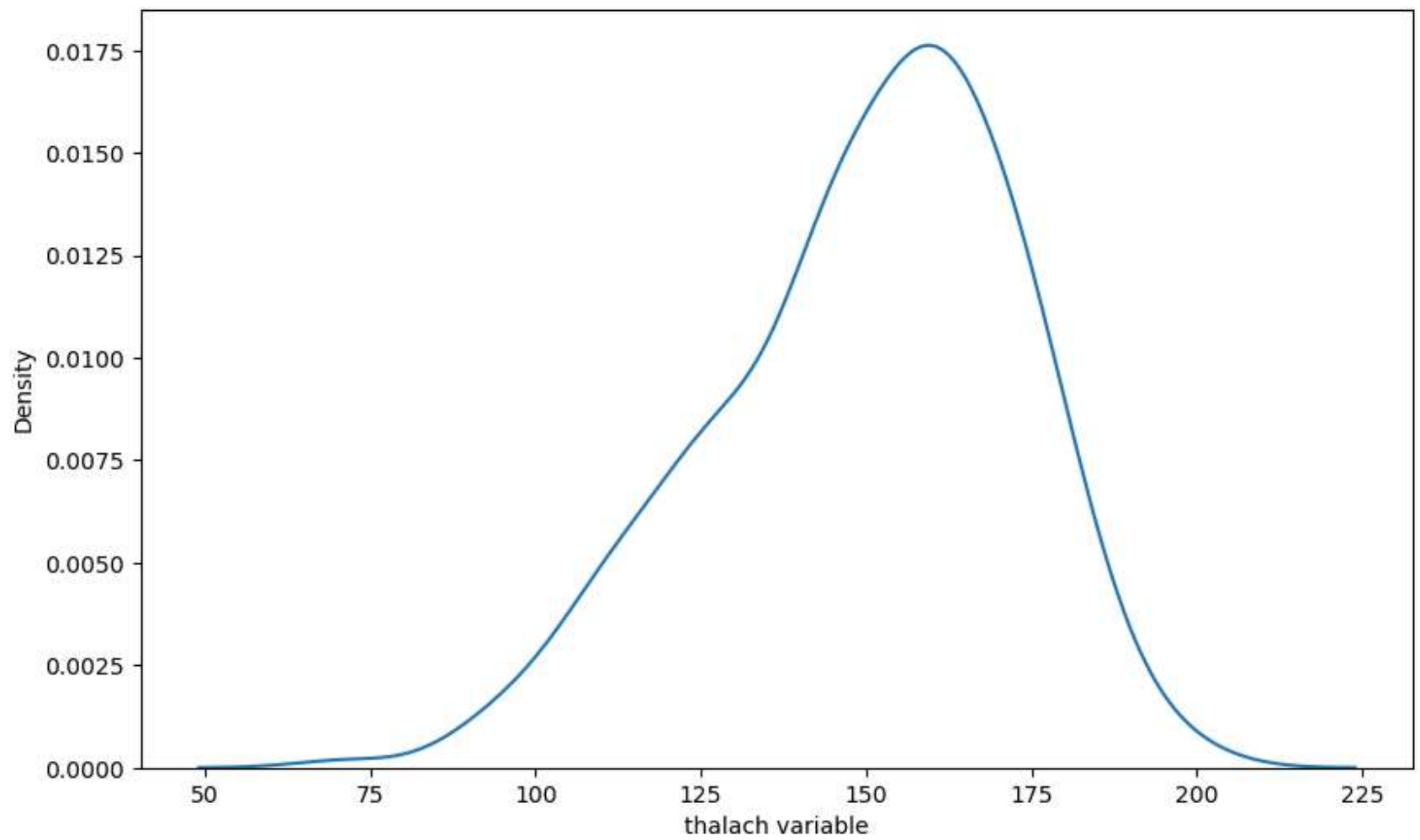
```
In [60]: f, ax = plt.subplots(figsize = (10,6))
x = heart['thalach']
x = pd.Series(x, name = "thalach variable")
ax = sns.distplot(x, bins = 10)
plt.show()
```



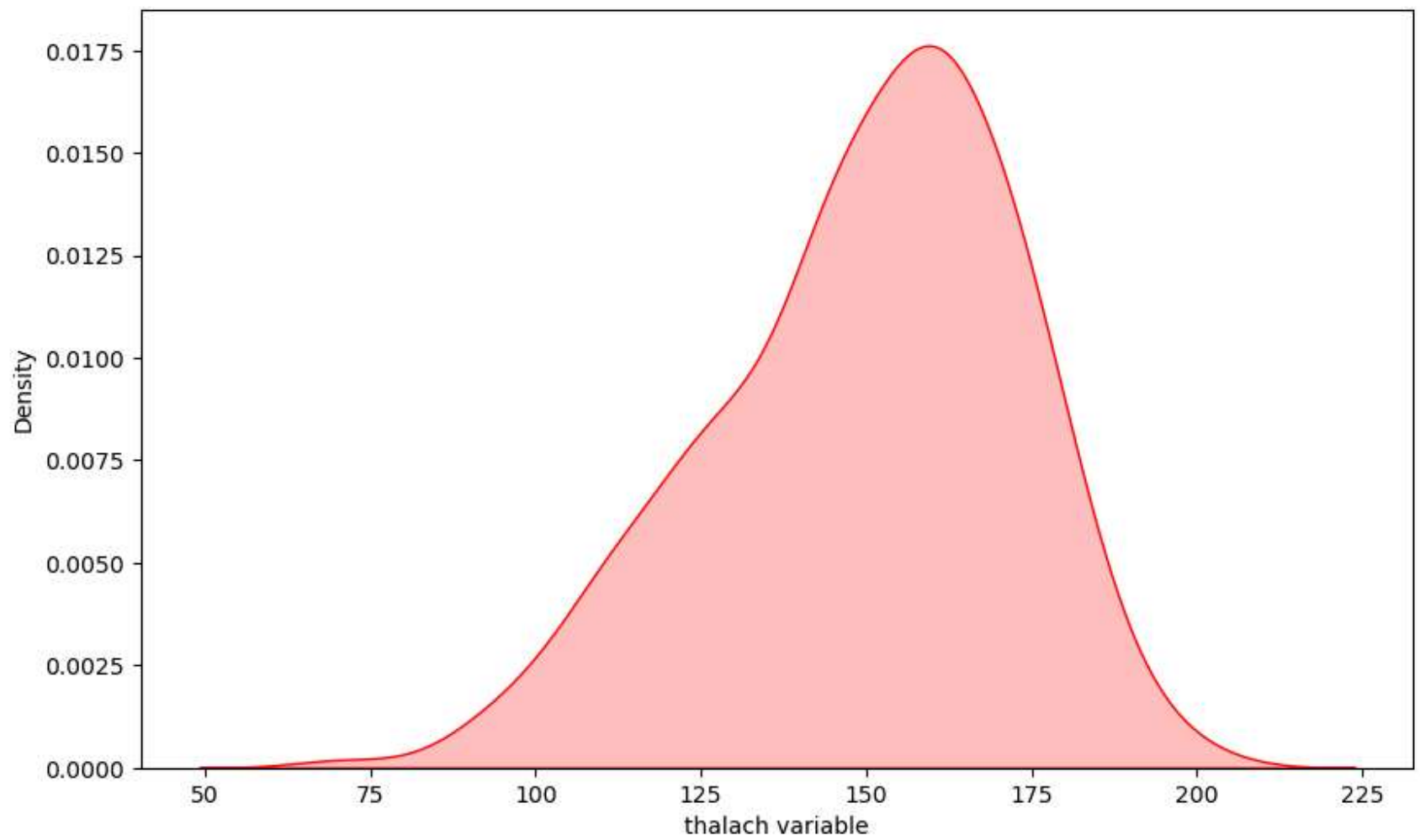
```
In [61]: f, ax = plt.subplots(figsize = (10,6))
x = heart['thalach']
ax = sns.distplot(x, bins = 10, vertical = True)
plt.show()
```



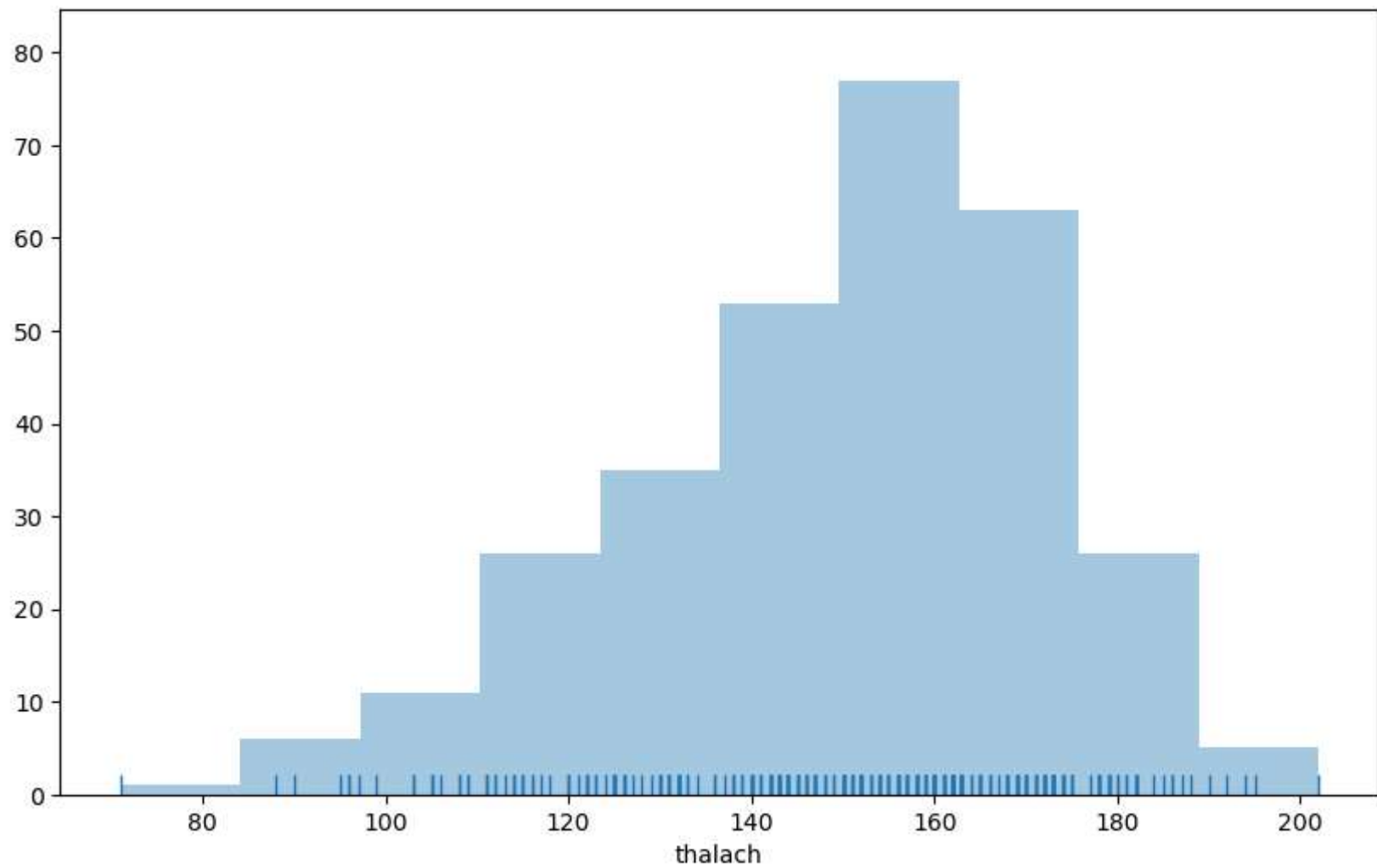
```
In [62]: f, ax = plt.subplots(figsize = (10,6))
x = heart['thalach']
x = pd.Series(x, name = "thalach variable")
ax = sns.kdeplot(x)
plt.show()
```



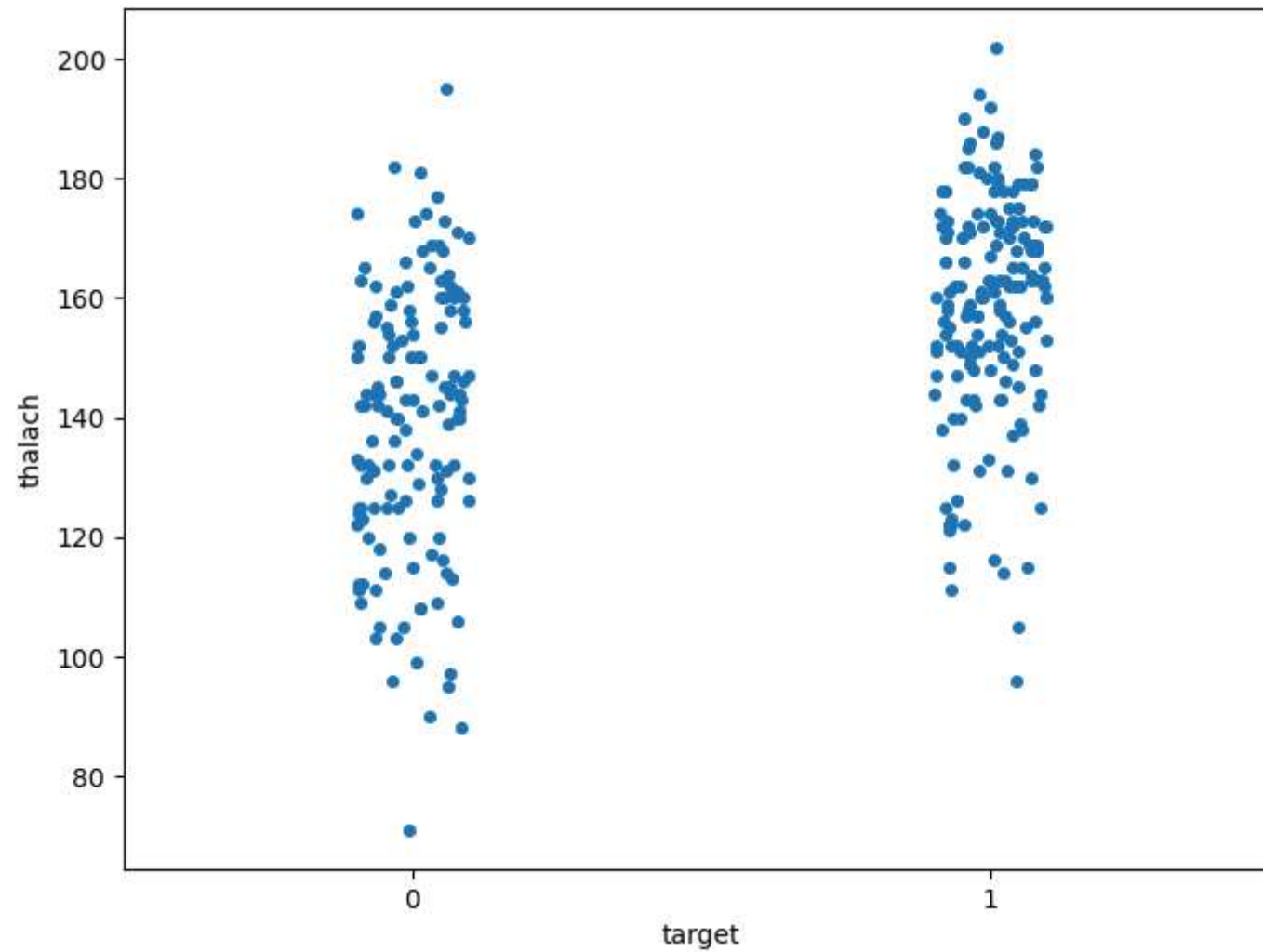
```
In [65]: f, ax = plt.subplots(figsize = (10,6))
x = heart['thalach']
x = pd.Series(x, name = "thalach variable")
ax = sns.kdeplot(x, shade = True, color = 'r')
plt.show()
```



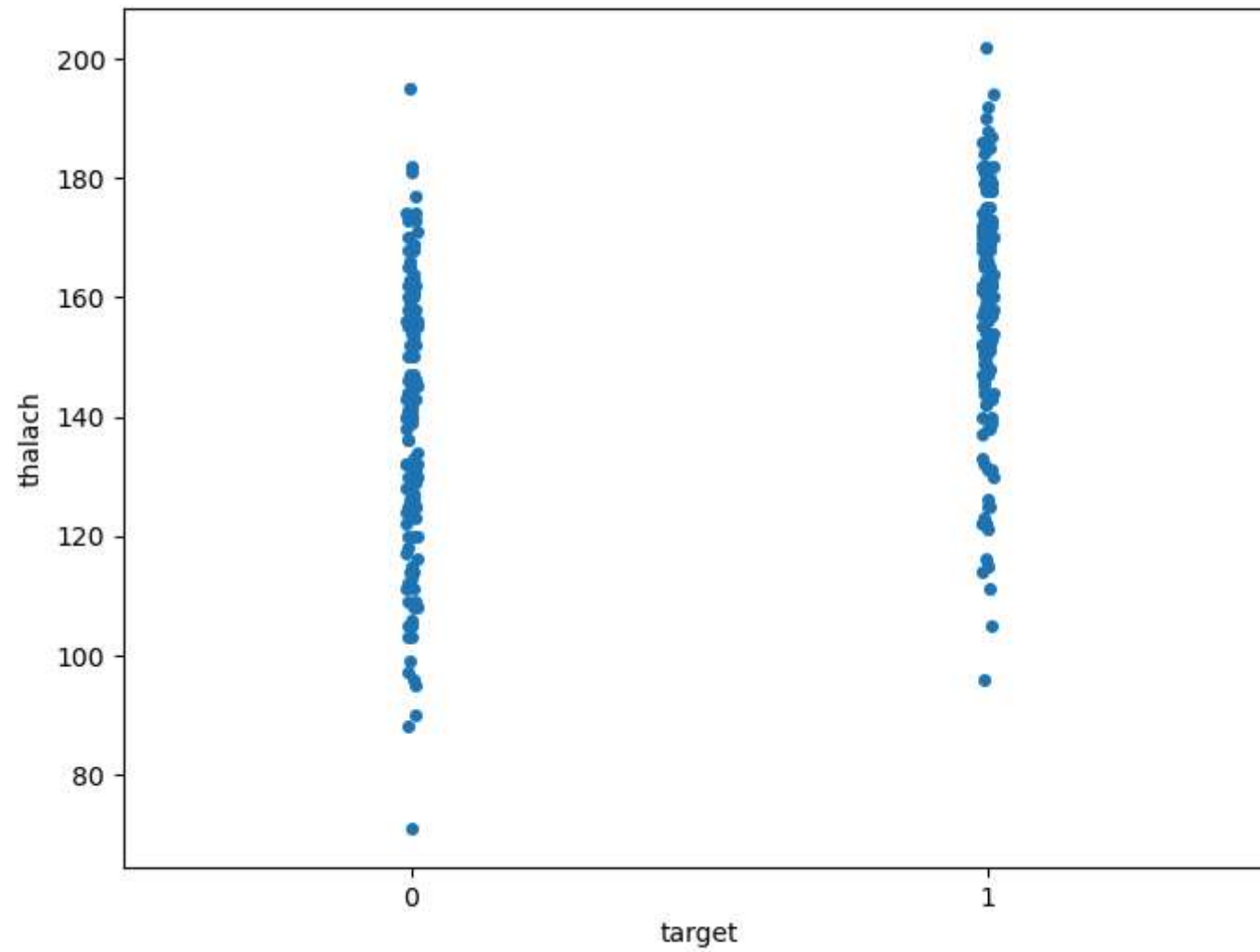
```
In [67]: f, ax = plt.subplots(figsize = (10,6))
x = heart['thalach']
ax = sns.distplot(x, kde = False, rug = True, bins = 10)
plt.show()
```

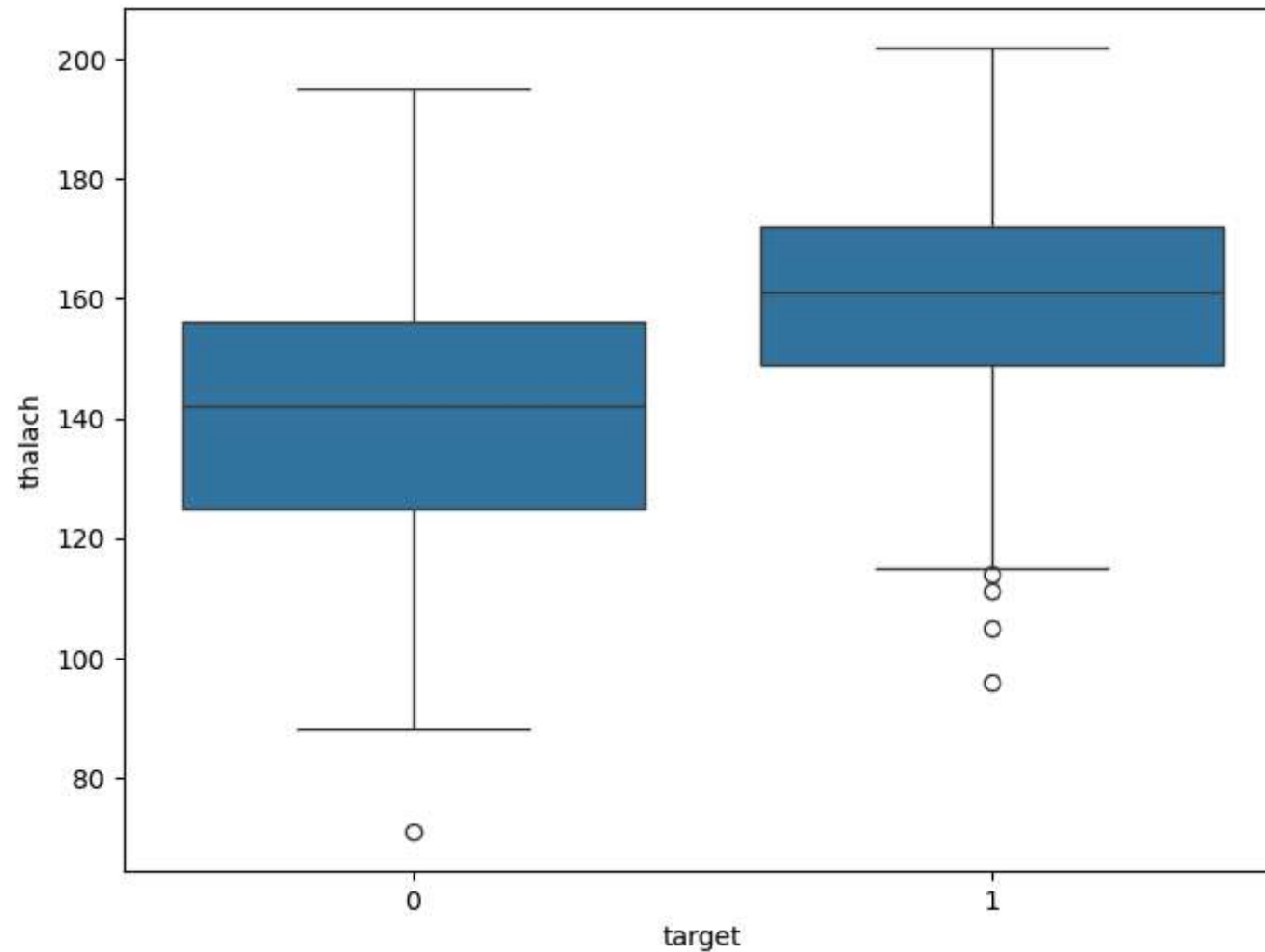
```
In [68]: f, ax = plt.subplots(figsize = (8,6))  
sns.stripplot(x = "target", y = "thalach", data = heart)  
plt.show()
```



```
In [69]: f, ax = plt.subplots(figsize = (8,6))  
sns.stripplot(x = "target", y = "thalach", data = heart, jitter = 0.01)  
plt.show()
```



```
In [70]: f, ax = plt.subplots(figsize = (8,6))
sns.boxplot(x = "target", y = "thalach", data = heart)
plt.show()
```



```
In [74]: plt.figure(figsize = (16, 12))
plt.title("Correlation Heatmap of Heart Disease Dataset")
a = sns.heatmap(correlation, square = True, annot = True, fmt = '.2f', linecolor = 'white')
a.set_xticklabels(a.get_xticklabels(), rotation=90)
a.set_yticklabels(a.get_yticklabels(), rotation=30)
plt.show()
```

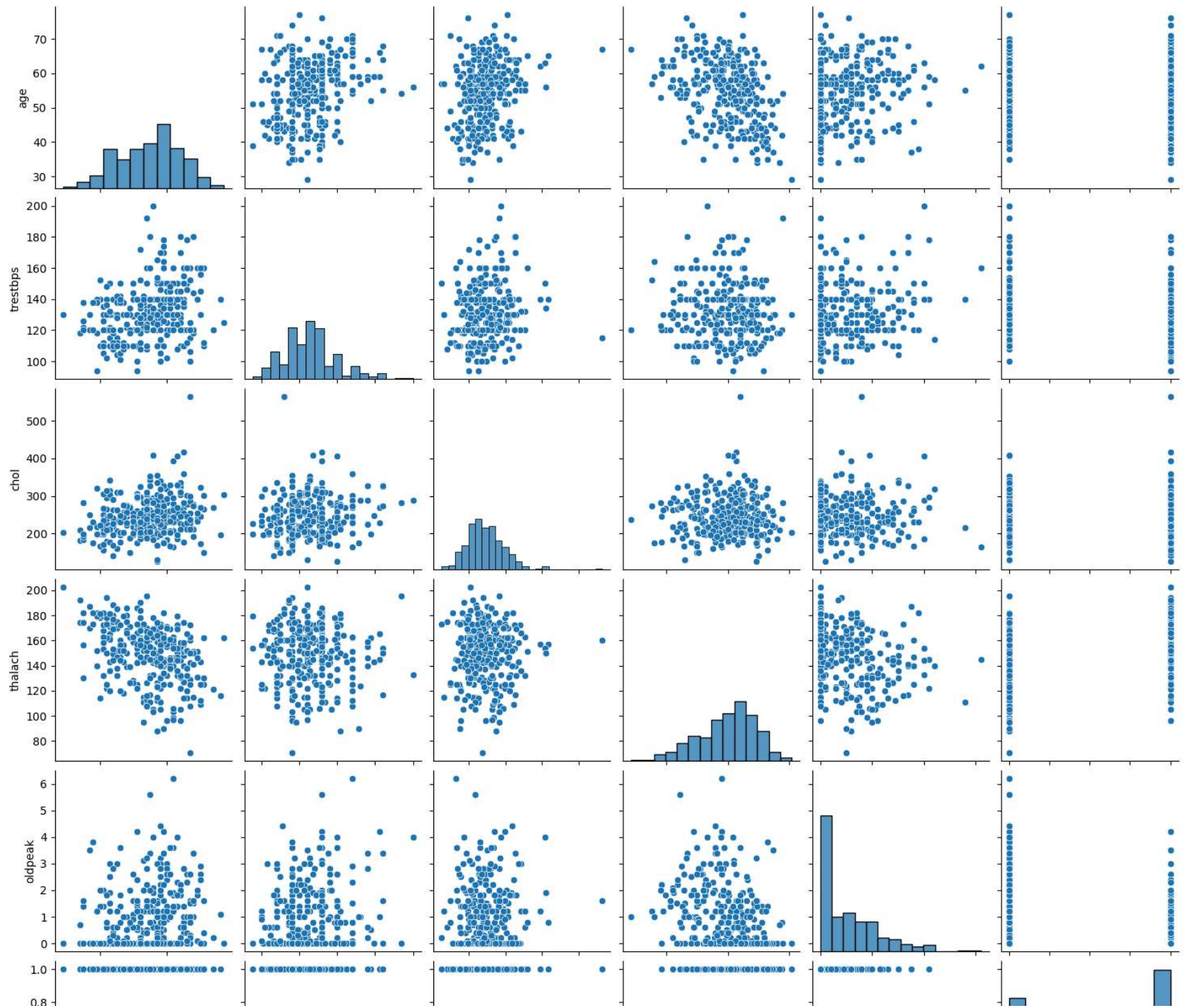


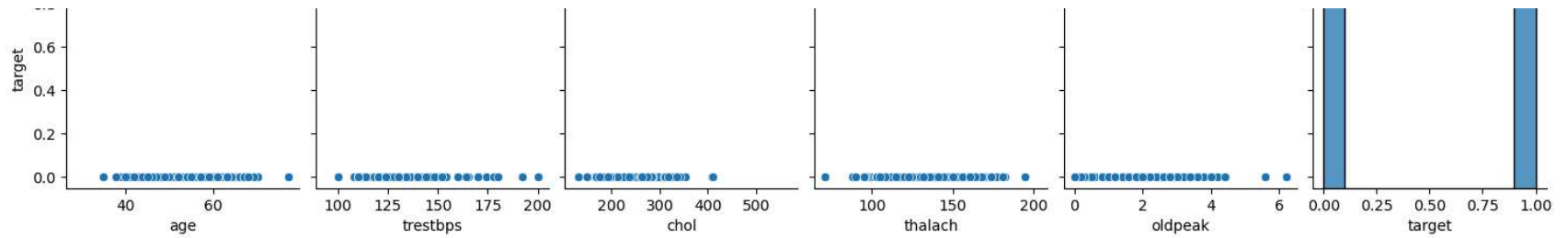
a s cp trestbps chol fbs restecg thalach exang oldpeak slope ca thal target

```
In [75]: heart.columns
```

```
Out[75]: Index(['age', 'sex', 'cp', 'trestbps', 'chol', 'fbs', 'restecg', 'thalach',  
              'exang', 'oldpeak', 'slope', 'ca', 'thal', 'target'],  
              dtype='object')
```

```
In [78]: num_var = ['age', 'trestbps', 'chol', 'thalach', 'oldpeak', 'target']  
sns.pairplot(heart[num_var], kind = 'scatter', diag_kind = 'hist')  
plt.show()
```





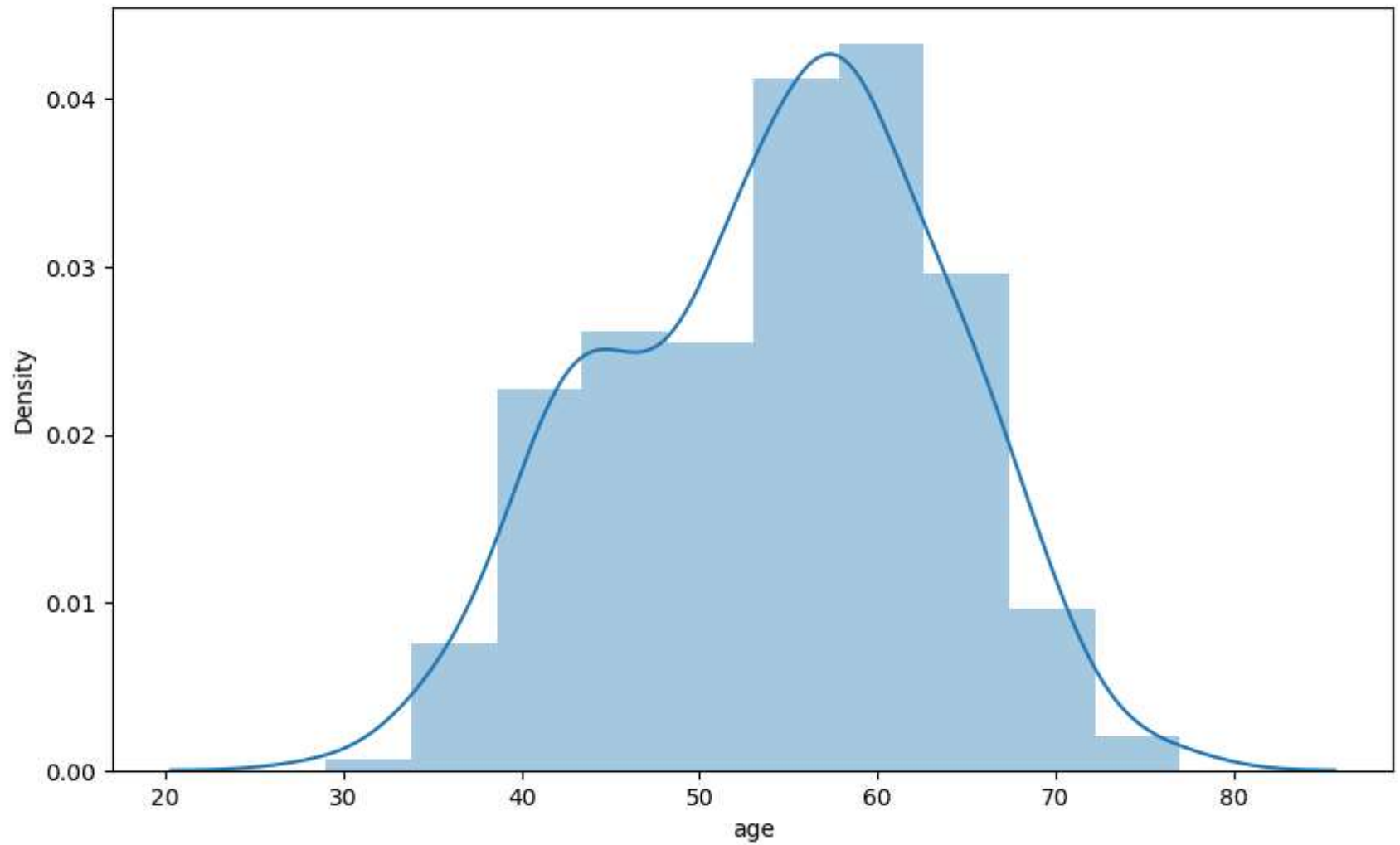
```
In [80]: heart['age'].nunique()
```

```
Out[80]: 41
```

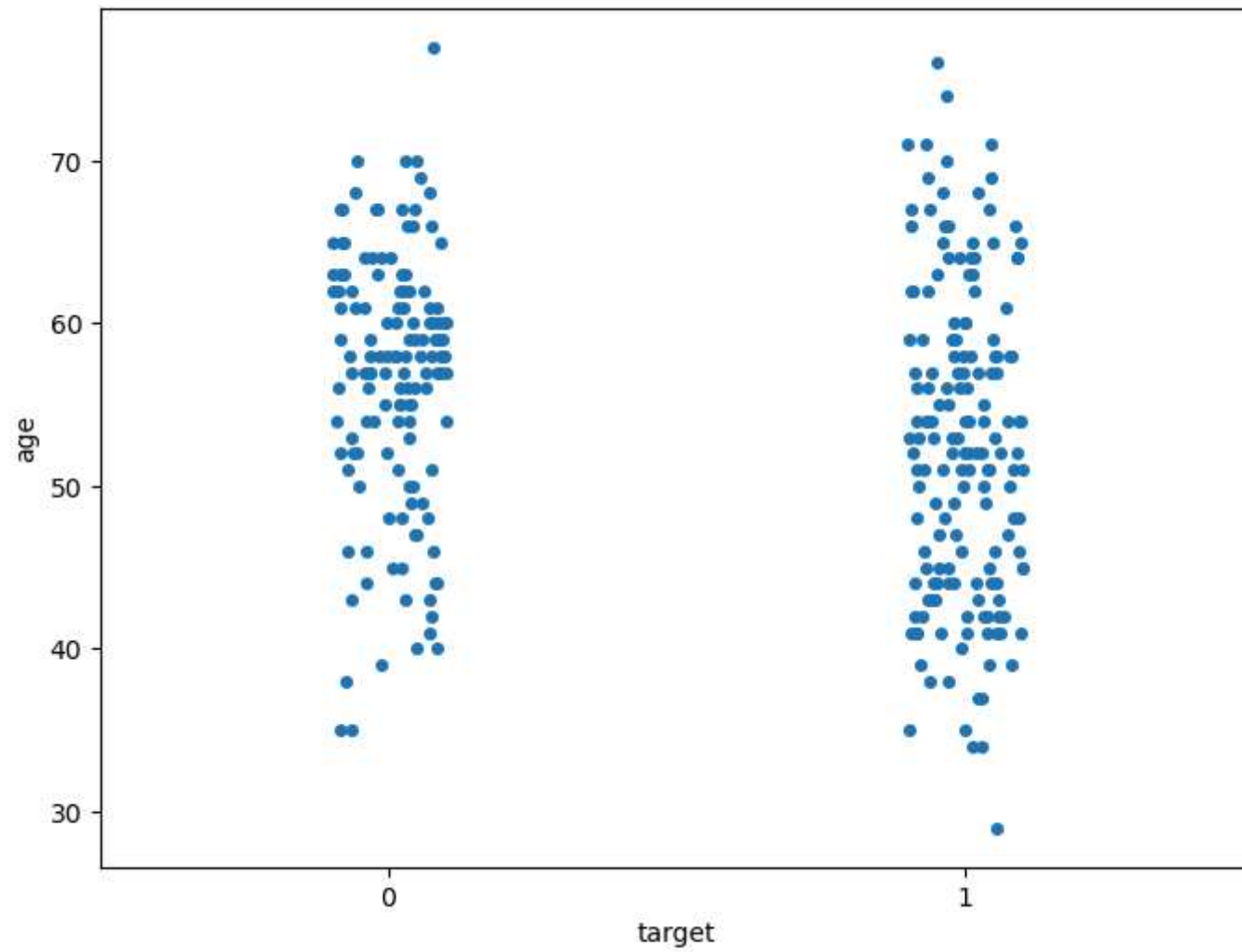
```
In [81]: heart['age'].describe()
```

```
Out[81]: count    303.000000
         mean      54.366337
         std       9.082101
         min       29.000000
         25%       47.500000
         50%       55.000000
         75%       61.000000
         max       77.000000
         Name: age, dtype: float64
```

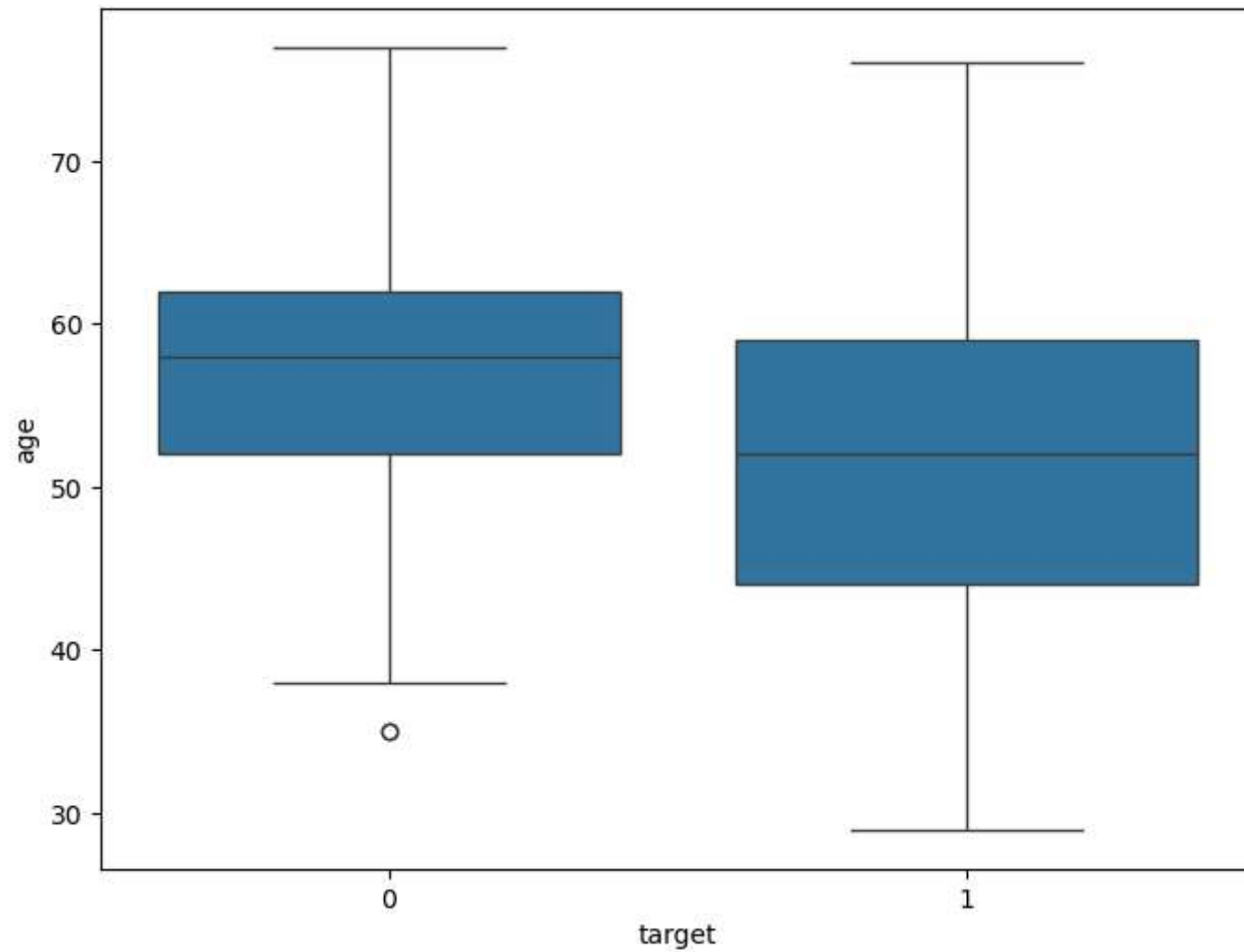
```
In [82]: f, ax = plt.subplots(figsize = (10,6))
         x = heart['age']
         ax = sns.distplot(x, bins = 10)
         plt.show()
```

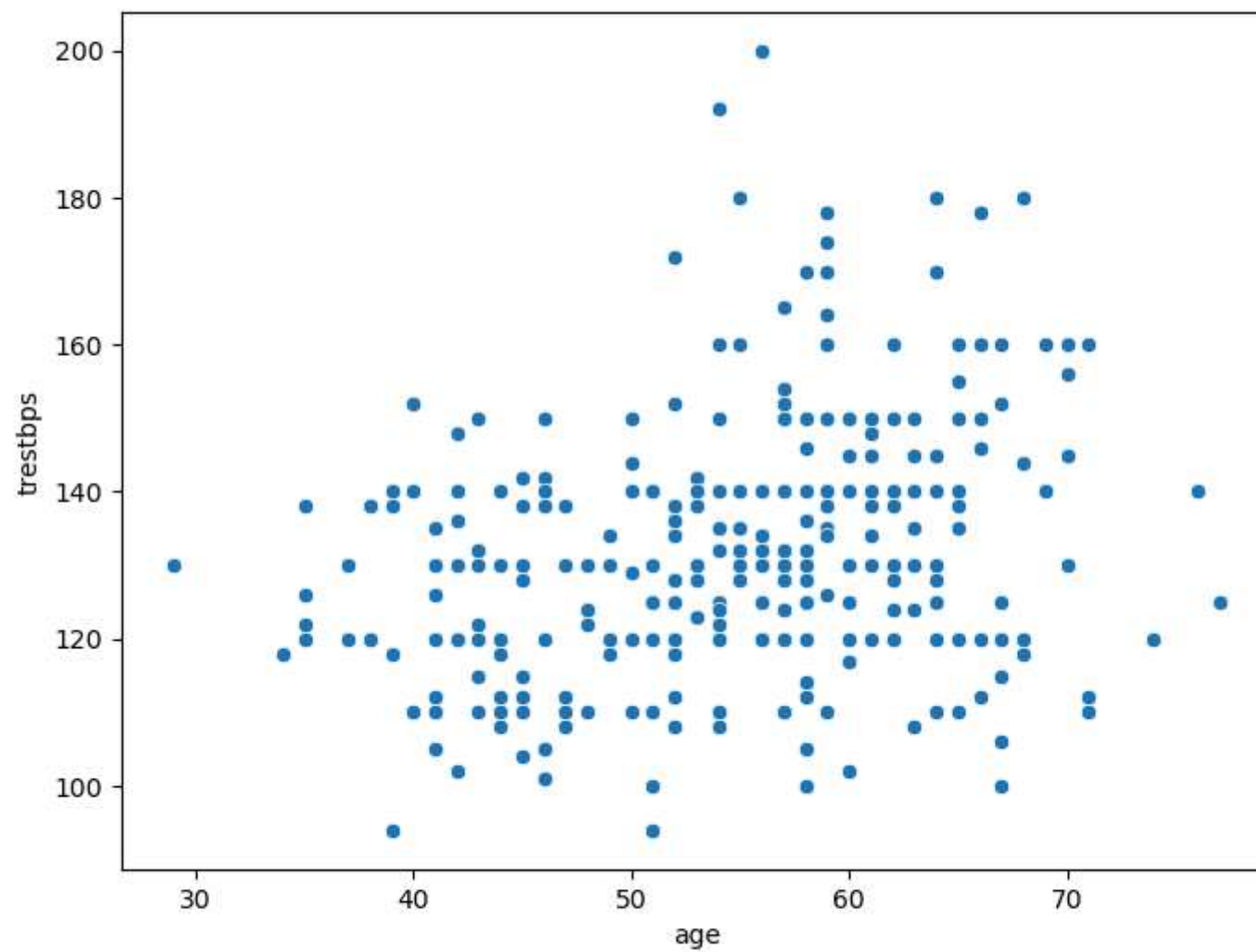
```
In [83]: f, ax = plt.subplots(figsize = (8,6))
ax = sns.stripplot(x = 'target', y = 'age', data = heart)
plt.show()
```



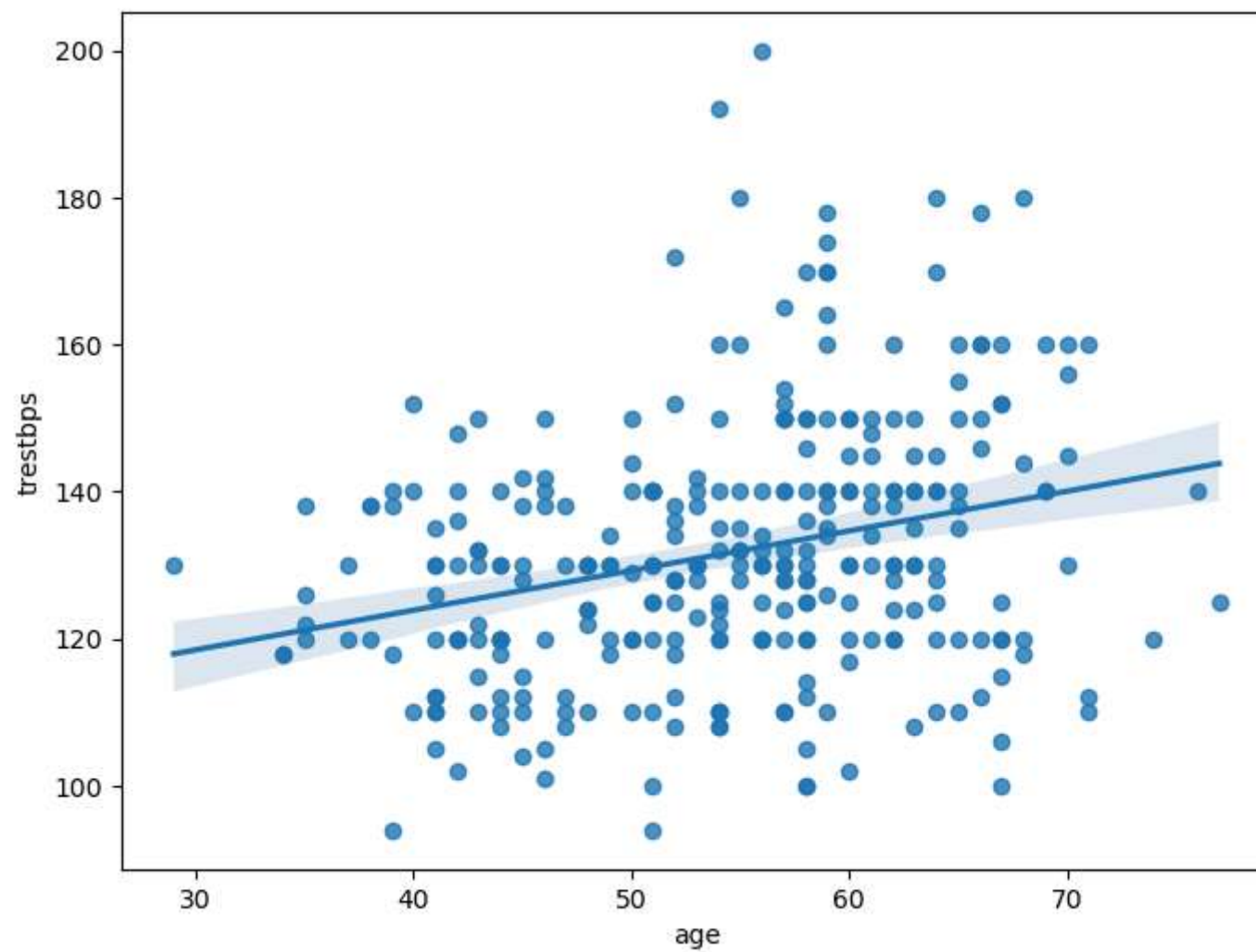
```
In [84]: f, ax = plt.subplots(figsize = (8,6))  
ax = sns.boxplot(x = 'target', y = 'age', data = heart)  
plt.show()
```



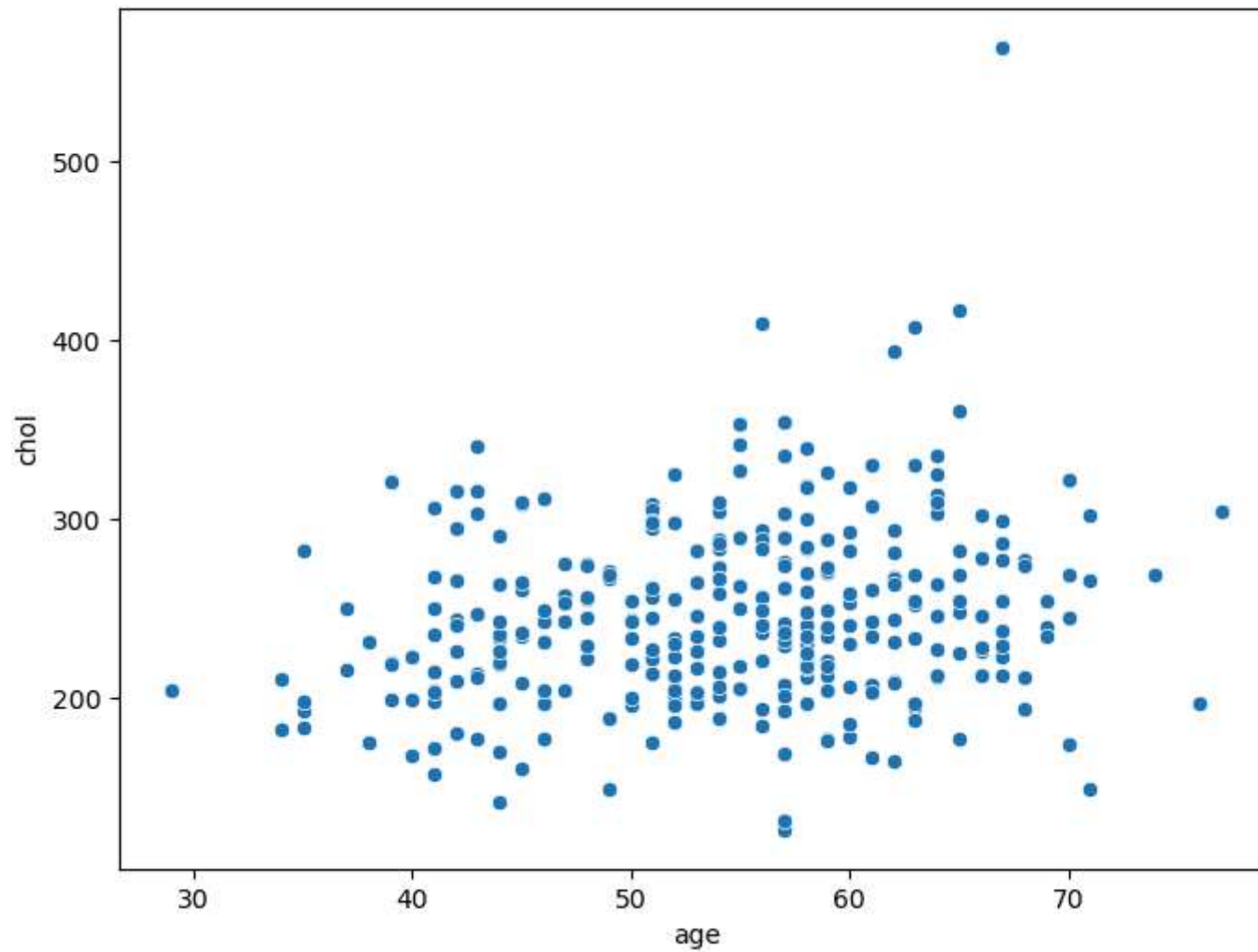
```
In [85]: f, ax = plt.subplots(figsize = (8,6))
ax = sns.scatterplot(x = 'age', y = 'trestbps', data = heart)
plt.show()
```



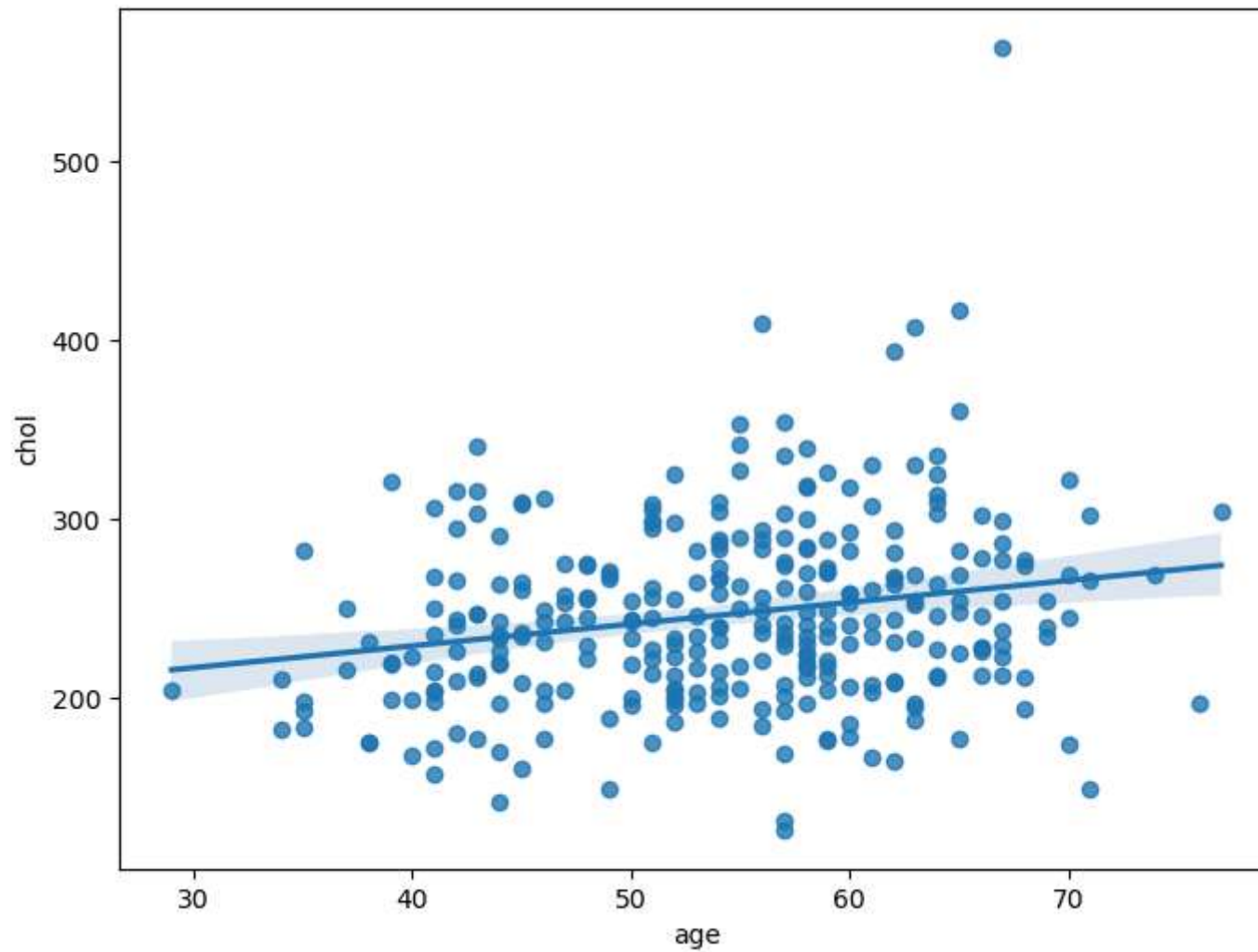
```
In [86]: f, ax = plt.subplots(figsize = (8,6))  
ax = sns.regplot(x = 'age', y = 'trestbps', data = heart)  
plt.show()
```



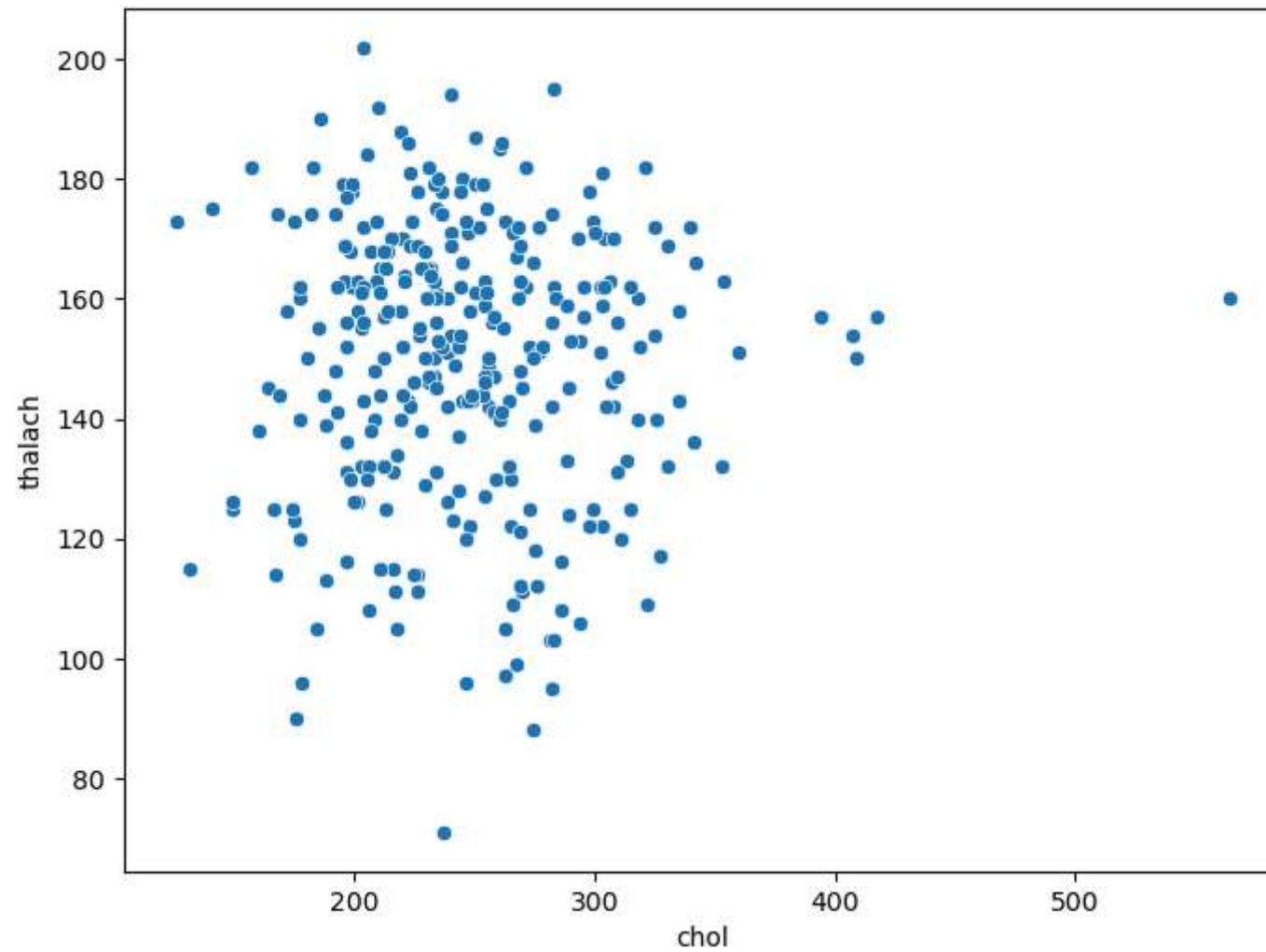
```
In [88]: f, ax = plt.subplots(figsize = (8,6))  
ax = sns.scatterplot(x = 'age', y = 'chol', data = heart)  
plt.show()
```



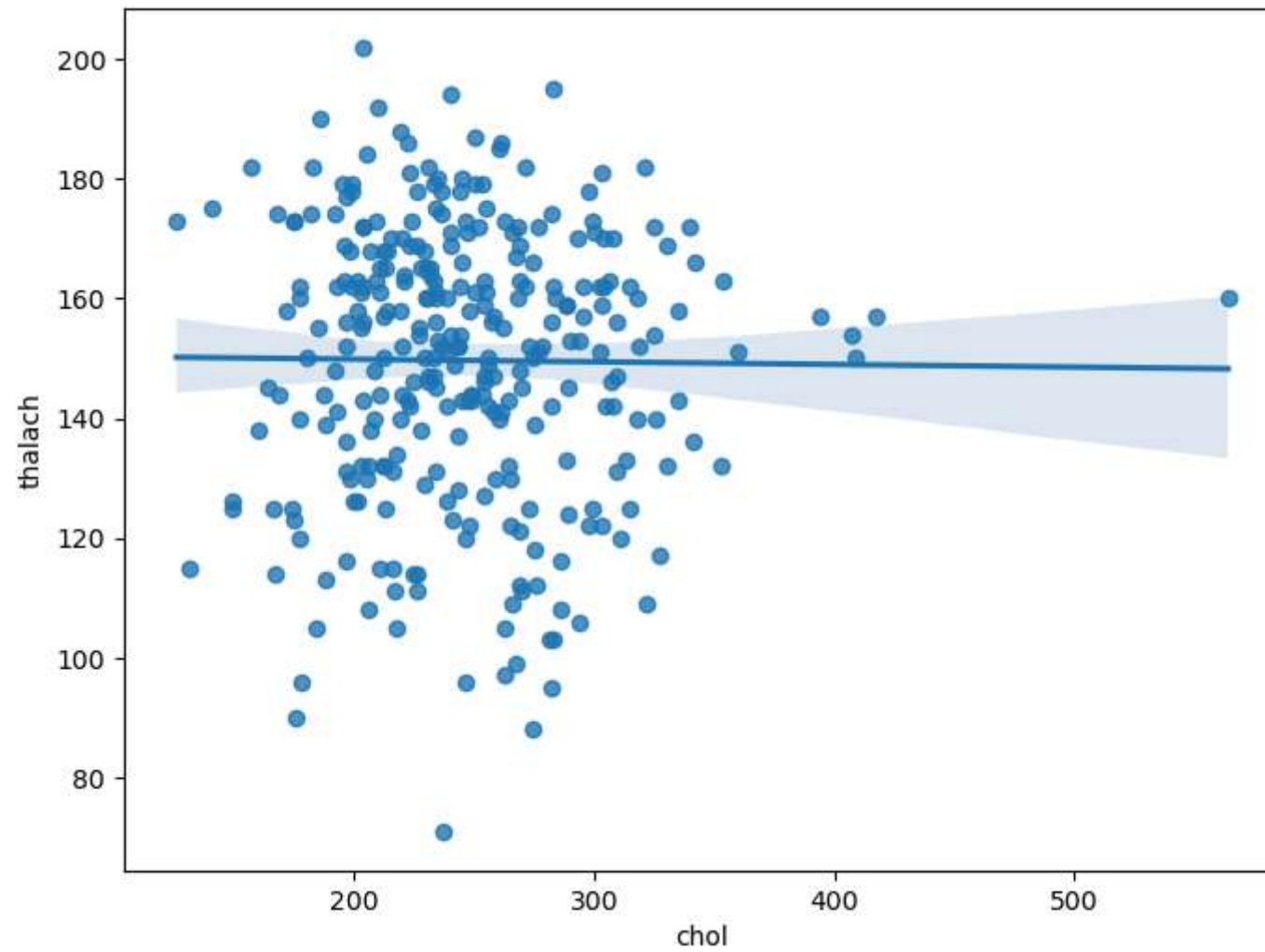
```
In [89]: f, ax = plt.subplots(figsize = (8,6))
ax = sns.regplot(x = 'age', y = 'chol', data = heart)
plt.show()
```



```
In [90]: f, ax = plt.subplots(figsize = (8,6))
ax = sns.scatterplot(x = 'chol', y = 'thalach', data = heart)
plt.show()
```



```
In [91]: f, ax = plt.subplots(figsize = (8,6))
ax = sns.regplot(x = 'chol', y = 'thalach', data = heart)
plt.show()
```

```
In [93]: heart.isna().sum()
```

```
Out[93]: age      0
sex      0
cp       0
trestbps 0
chol     0
fbs      0
restecg  0
thalach  0
exang    0
oldpeak  0
slope    0
ca       0
thal     0
target   0
dtype: int64
```

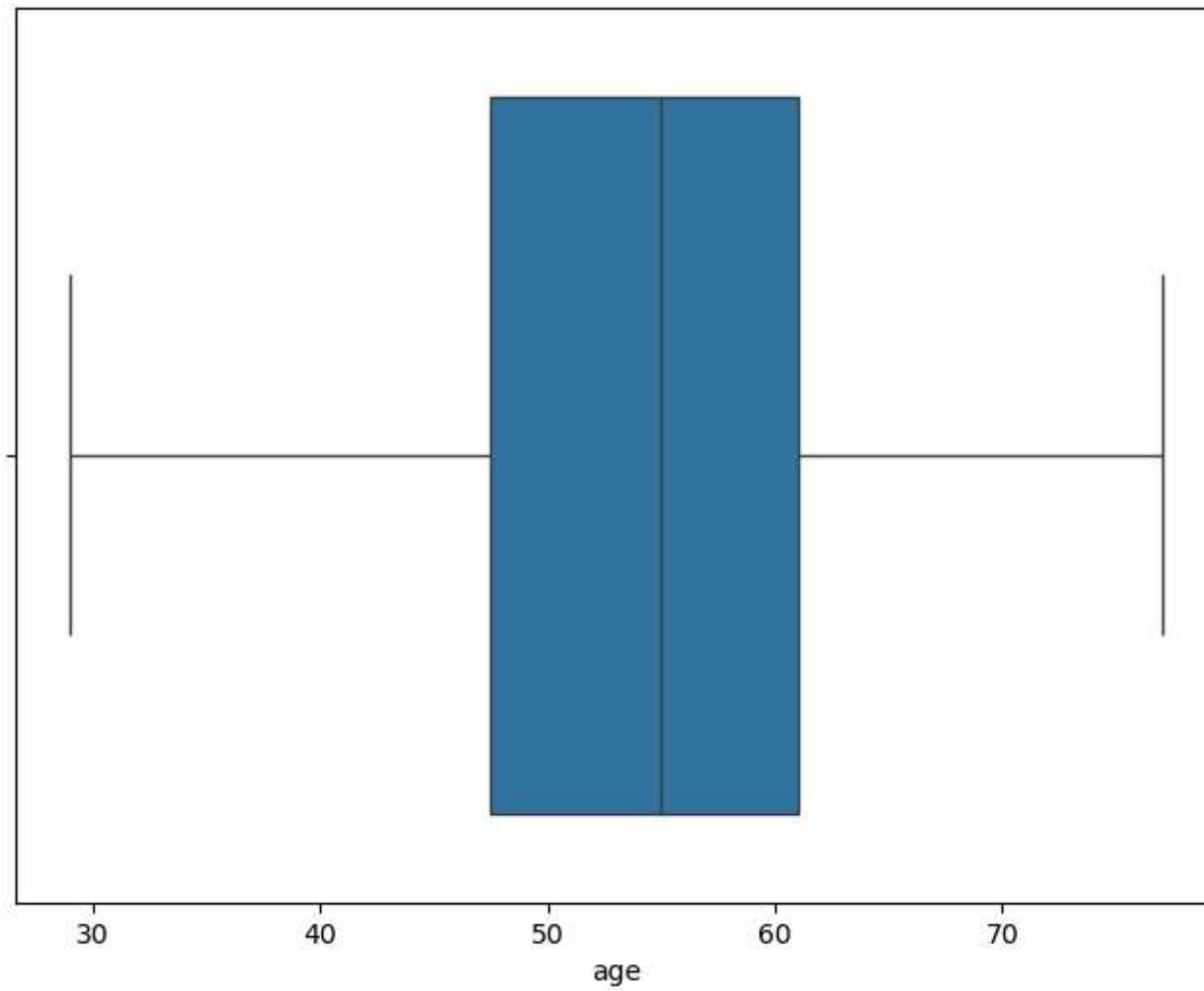
```
In [94]: assert pd.notnull(heart).all().all()
```

```
In [95]: assert (heart>=0).all().all()
```

```
In [96]: heart['age'].describe()
```

```
Out[96]: count    303.000000
mean      54.366337
std       9.082101
min       29.000000
25%      47.500000
50%      55.000000
75%      61.000000
max       77.000000
Name: age, dtype: float64
```

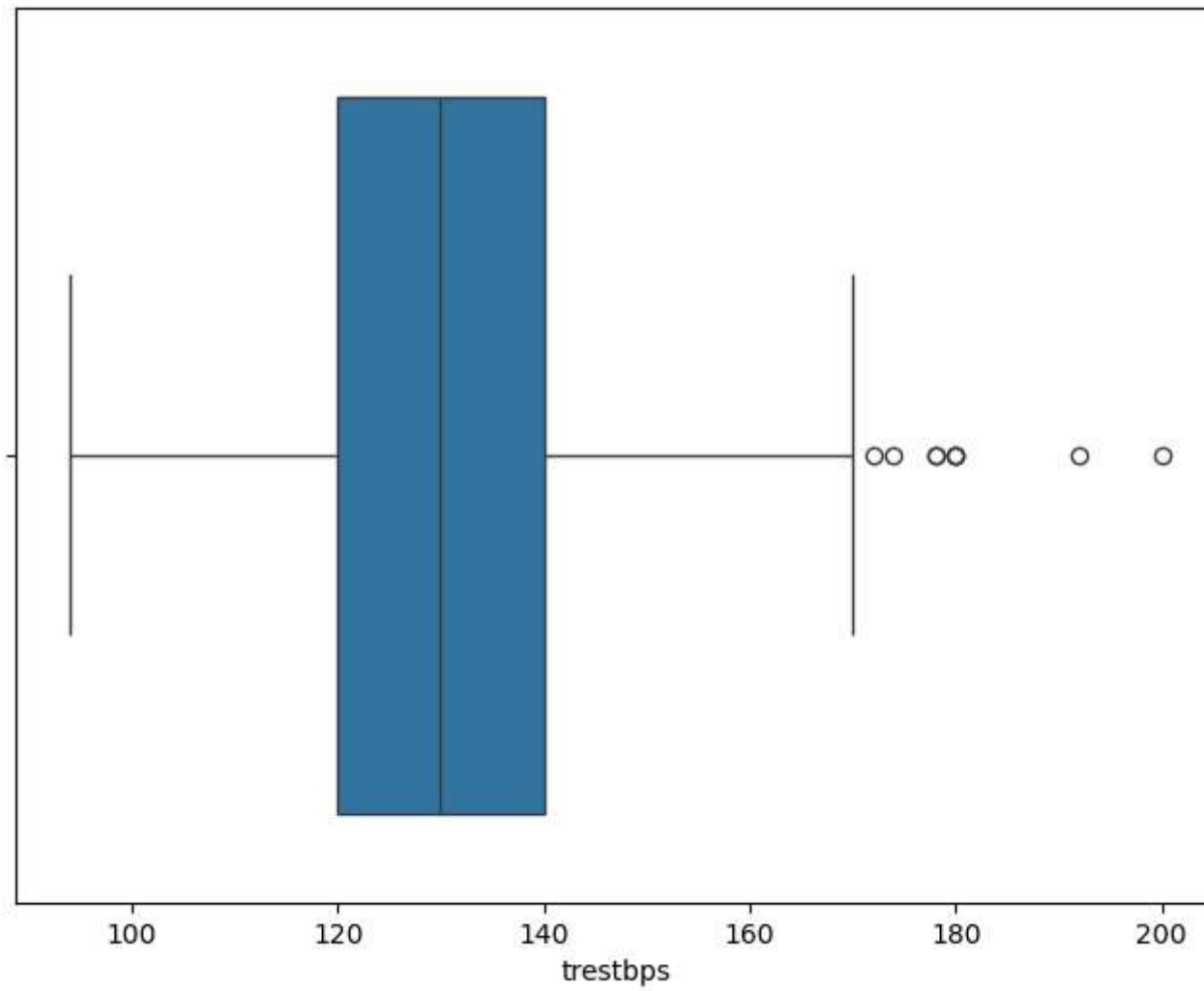
```
In [97]: x, ax = plt.subplots(figsize=(8,6))
sns.boxplot(x=heart['age'])
plt.show()
```



```
In [98]: heart['trestbps'].describe()
```

```
Out[98]: count    303.000000
          mean     131.623762
          std      17.538143
          min      94.000000
          25%     120.000000
          50%     130.000000
          75%     140.000000
          max     200.000000
          Name: trestbps, dtype: float64
```

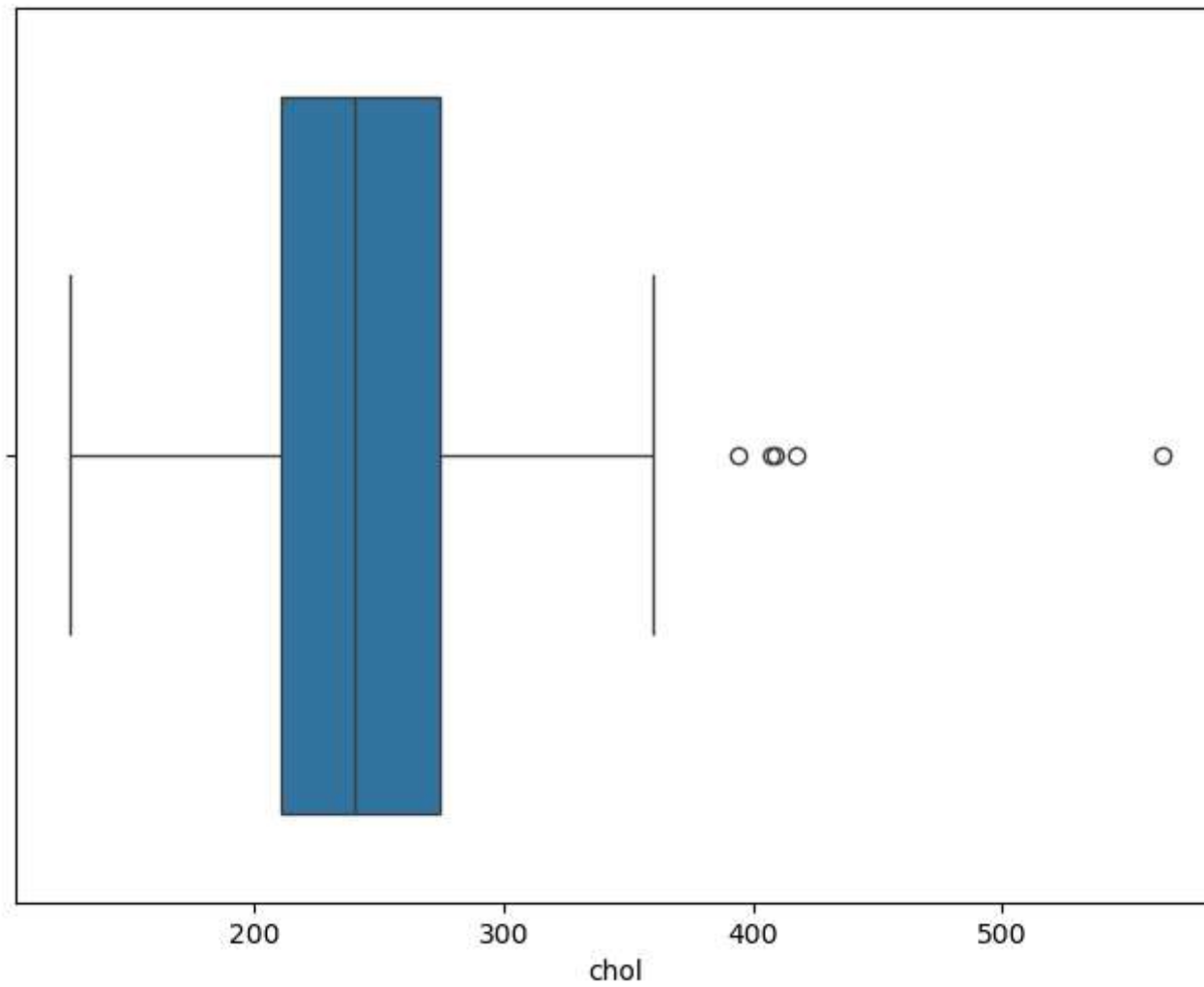
```
In [99]: x, ax = plt.subplots(figsize=(8,6))
          sns.boxplot(x=heart['trestbps'])
          plt.show()
```



```
In [100... heart['chol'].describe()
```

```
Out[100... count    303.000000
          mean    246.264026
          std      51.830751
          min     126.000000
          25%     211.000000
          50%     240.000000
          75%     274.500000
          max     564.000000
          Name: chol, dtype: float64
```

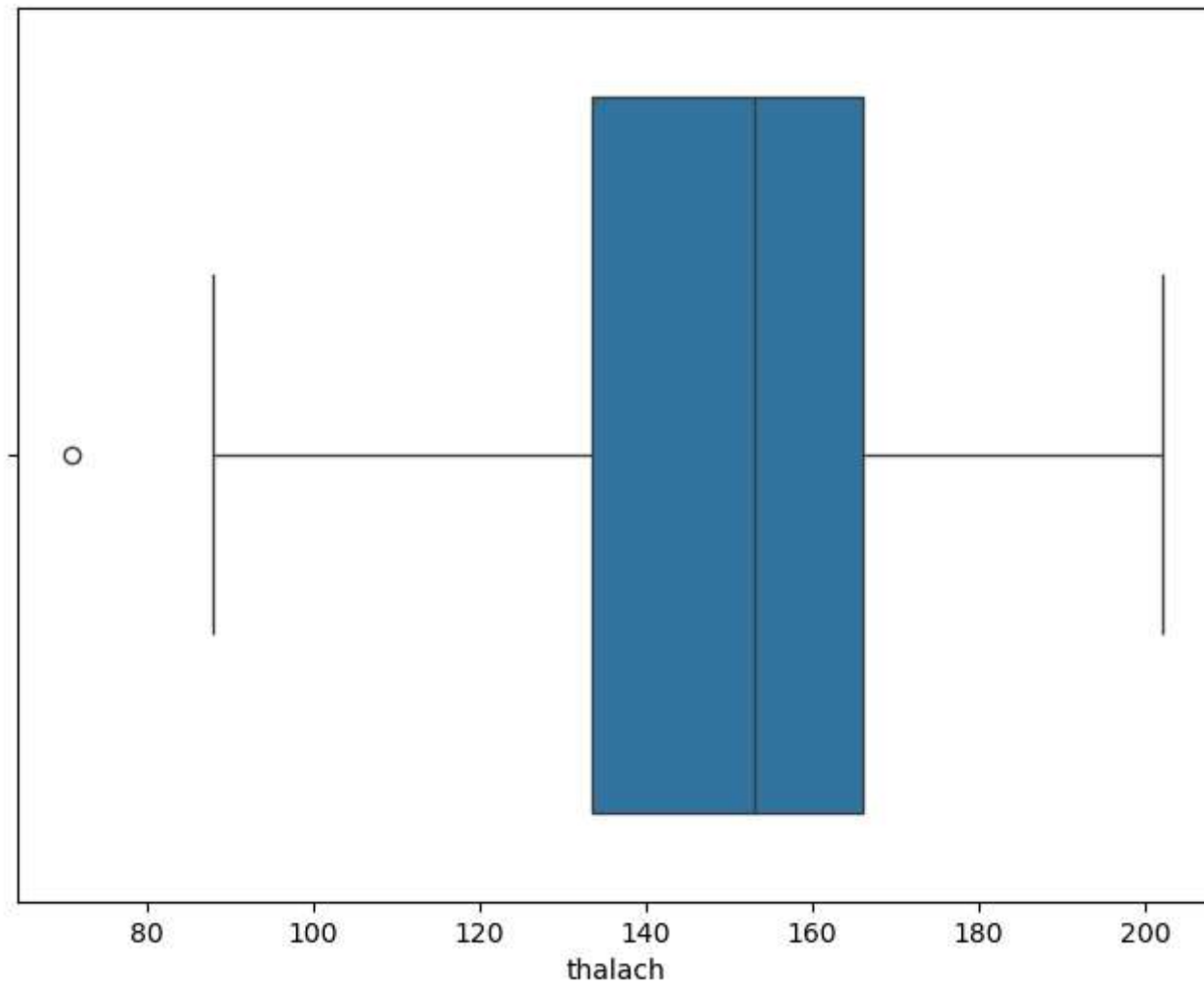
```
In [101... x, ax = plt.subplots(figsize=(8,6))
          sns.boxplot(x=heart['chol'])
          plt.show()
```



In [102... `heart['thalach'].describe()`

```
Out[102... count    303.000000
          mean    149.646865
          std     22.905161
          min     71.000000
          25%    133.500000
          50%    153.000000
          75%    166.000000
          max     202.000000
          Name: thalach, dtype: float64
```

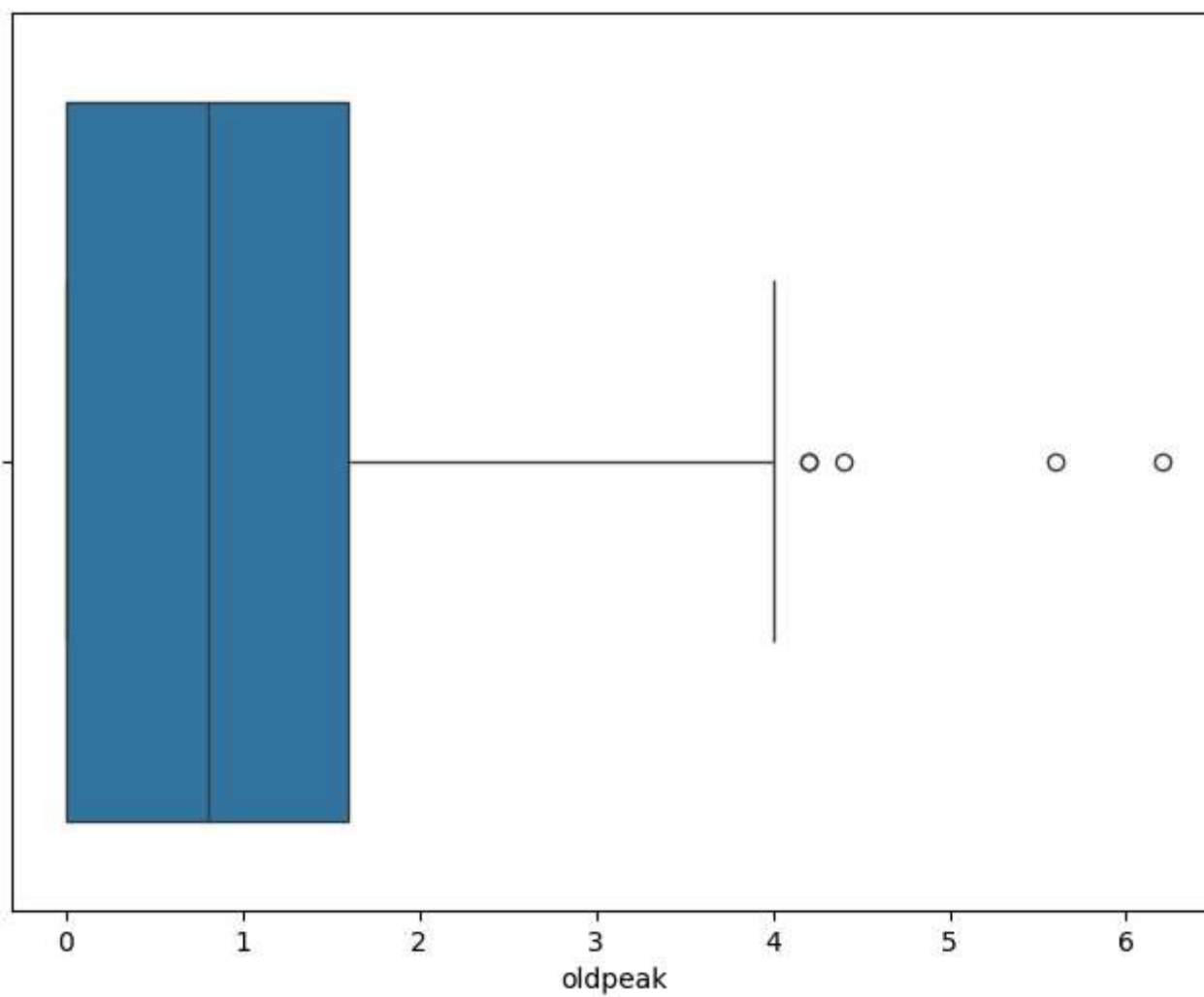
```
In [103... x, ax = plt.subplots(figsize=(8,6))
          sns.boxplot(x=heart['thalach'])
          plt.show()
```

In [107... `heart['oldpeak'].describe()`

```
Out[107... count    303.000000
mean      1.039604
std       1.161075
min       0.000000
25%      0.000000
50%      0.800000
75%      1.600000
max       6.200000
Name: oldpeak, dtype: float64
```

```
In [108... x, ax = plt.subplots(figsize=(8,6))
sns.boxplot(x=heart['oldpeak'])
plt.show()
```



In []: